

Local Labour Markets and Skill Acquisition

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Abstract

Regional inequalities in developed economies have been the subject of both a large academic literature and policy interest. Yet, little is known about the causal impact of local skill demand on the supply of skills by local students. I study this relationship in England, which has large spatial disparities; an educational system where local economic conditions have little direct impact on school funding; standardized test scores; and detailed data on outcomes by field. I use establishment-level administrative data to document the cross-sectional and panel variation in the level and field of skills demanded. I then combine this with individual-level education data to study whether local skill demand shapes local students' educational attainment: their results; level of education; and fields of study. I document a positive cross-sectional correlation between the skills demanded in local jobs and education choices. But, using a dynamic difference-in-difference strategy, I find at most a very muted response to large increases in local demand for degrees or specific skills for subsequent cohorts of students making educational investment decisions. I discuss the implications of my findings for policies aiming to target regional inequality.

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1 Introduction

There are large and persistent differences in educational attainment and skills across locations within countries. These spatial inequalities have recently drawn increased attention for their potential economic and political impacts, as well as for the individual consequences for those in disadvantaged areas. However, more insight is needed as to the ways in which various local characteristics are perpetuated. One important decision facing young people is the choice of education - both in terms of the type and level of education they pursue and the field in which they study.

Within England there are large, persistent, differences in educational attainment across areas, which have garnered significant concern and media attention. The gaps between London and rural areas have been documented at least as far back as the mid-nineteenth-century (White, 2007) and have persisted into the twenty-first century (Overman and Xu, 2022). Despite England’s relatively small size, location is sticky: 65% of university graduates and 85% of non-graduates live in the same Travel to Work Area (TTWA) in which they grew up (Britton et al., 2021). These characteristics lead to a natural research question: How responsive are students’ educational investment decisions to local skill demand in the labour market in which they grow up?

While I study this question in the context of England (for reasons discussed below), this setting is not unusual in its spatial disparities. For example, in the US, the share of adults with university degrees ranges from less than 27% in Louisiana and Mississippi to more than 42% in Massachusetts and Vermont, and household income in the poorest states is 55% of that in the richest (U.S. Census Bureau, 2019). And despite overall higher (though declining) geographic mobility, 60% of Americans live in the same state in which they were born Jia et al. (2023).

Governments have aimed to address spatial inequalities in part through place-based industrial policy, using both government jobs and incentives for private investment. In the UK, these policies have included the Places for Growth scheme, enacted, in part, to “[ensure] economic growth and job opportunities are more evenly distributed across the UK” by moving more than 20,000 Civil Service jobs outside of London (Department for Business,

2021). In the US, the Helping Infrastructure Restore the Economy (HIRE) Act, proposed in 2019, would have required the relocation of several federal government agencies away from Washington, D.C.. More recently, the Creating Helpful Incentives to Produce Semiconductors (CHIPS) and Science Act of 2022 provided \$39 billion of funding to support local hubs and foster the growth of the semiconductor industry.

Understanding whether such policies can spur local skill accumulation is important. The most direct route by which demand may affect locals' skills is through cohorts of students who may change their skill acquisition in response. In the case where the elasticity of education choices to the jobs available is small, these policies may instead primarily change local skill composition through (costly) moves of workers from other locations.

To better understand how students' educational investment decisions respond to local skill demand in the labour market in which they grow up, I turn to England. In addition to the spatial disparities described above, England has many advantageous institutional features for this question. Firstly, school funding is determined nationally on a per-pupil basis, meaning an expansion in local employment or an increase in house prices does not increase education funding. This stands in sharp contrast to other contexts where local taxes provide the bulk of education funding, confounding estimates of student response¹. Furthermore, both academic and vocational qualifications fit into a standardized framework from before the end of compulsory schooling, allowing me to compare more specific skills at an earlier stage of education. The comparability of qualifications across time and space, as well as comprehensive data on student results, allows me to examine the field of study, student achievement, and level of education. Finally, administrative data on students and firms is available to study.

In this paper, I use rich administrative data on both establishments and students in England to evaluate the education response of students to changes in the composition of local skill demand. I examine the role that existing differences in skill demand play in local students' educational attainment: their results; level of education; and fields of study. I then exploit large changes in local demand for skills - either qualifications in a specific field

¹Consider an influx of well-paid jobs requiring degrees; the resulting increase in local property tax base would mechanically increase school funding, an important contributor to school quality, which is associated with higher levels of educational attainment (Eckert and Kleineberg, 2021).

or at the degree level - to measure the short- to medium-run response of local students. These changes come from changing local industry composition; I show that, following a large change, local demand for that skill remains at a significantly higher level than it was in the preceding years.

To understand local skill demand, I use the Business Structure Database (BSD), an administrative dataset covering an estimated 99% of economic activity in the UK. This establishment-level data provides information on the precise location, industry and employment of each establishment. I combine this with survey data on the qualifications held by employees in each sector to measure the mix of skills (by level and field), used in each labour market. The granular nature of the data allows me to consider other geographic levels, as well as changes coming from specific events.

On the student side, I use English administrative data on the universe of students who attended state (publicly funded) schools. This data, which is linked to university records, allows me to observe the full educational path - including the subjects studied and results - of students from primary through tertiary education.

My primary unit of analysis is the TTWA—these regions are defined by the ONS using census data such that the bulk (at least 75%, or 66.7% in larger areas) of their resident population work within the same area. These regions are mutually exhaustive and collectively exclusive, covering the whole of the UK. Conceptually, they are similar to Commuting Zones and can be thought of as local labour markets². Thus, the type of employment present in the TTWA can be thought of as characterising the labour market which students in an area will face. In some specifications, I also consider results at the Postcode District (PD) level. This smaller unit is roughly equivalent to a zip code; it has an average area of 25 square miles.

In the first part of the paper, I document the spatial differences in industries and their associated skills and descriptively relate them to the spatial differences in education attainment. In this section, I show that students' education choices, in both level and field, are strongly correlated with the estimated skill distribution of local employment. I then move

²While recent work (e.g. Manning and Petrongolo (2017), Monte et al. (2018)) emphasises the importance of local spatial linkages, this would be better incorporated using LEO data to empirically understand the expected labour market for these cohorts

to a staggered event study design exploiting large changes in local skill demand to identify causal estimates of students' response.

First, I look at how the estimated share of local jobs requiring a degree relates to student enrolment. I find that, in the cross-sectional correlations, compared to areas in the 25th percentile of share of degree jobs, those in the 75th have 10pp more students enrolling in A-levels (compared to a national average of 45%) and 8pp more students enrolling in university (compared to a national average of 27%). I then consider the choice of field of study in university; here, for every additional 10pp of local employment in a field (compared to other TTWA), there is a 3pp difference in university enrolment in that field.

I then exploit large changes in local area skill composition to understand whether and when students respond to changing local conditions. I consider 1-year increases of more than 1pp in local degree employment in the TTWA where a student began secondary school; these changes are persistent. Despite the strong correlational relationships, I find no significant changes in enrollment in A-Levels or University following an increase in degree employment, nor change in field enrolment following an expansion of local employment in that field. To bound the response, I turn to students' immediate surroundings and consider the same change at the postcode group level. Here I also find no evidence for an increase in A-level enrolment, and can bound the response from above by a 2.5pp increase in enrolment for every 1pp increase in the share of jobs requiring a degree. The response for university enrolment is similarly small with the upper bound at 1.8pp for the same increase; this translates to just 5 additional students enrolled in university for every 100 new jobs requiring degrees.

In response to local increases in field share of more than 0.75% of local employment, I first show that these expansions are persistent - the level of employment in that field in the subsequent years remains significantly above its initial level. I then examine the student response. At the TTWA level, I find no evidence of increased field enrolment. Again turning to the postcode group level, I can bound the response from above at a 1pp increase in student enrolment for every 1.5pp increase in field employment.

Given the strong correlation between the skills used in local jobs and students' education choices, the small short-run response of students to changing circumstances warrants further consideration. It could be the case that the period of adjustment is longer than I observe in

my data; for example, if local teachers know more about the pathways that previous cohorts of students took, or if students beliefs about the labour market they will enter update slowly. It could also be the case that students do not view local changes as relevant for their later job prospects.

The muted response of local students has important implications for policies targeting regional inequalities. The magnitude of responses I find are not economically meaningful: the response is not enough to fill newly-created high-skill jobs with local students. If that is the goal of these policies, additional intervention may be needed to shape education choices. However, an important caveat is that my results relate only to academic qualifications; it is also relevant to understand whether other skills, such as those requiring vocational training, which are more likely to be used locally are likely to respond to changes.

This also has implications for the impact of these policies on local growth. If local students do not fill the demand, the vacancies created by programs intended to move high-skill jobs will need to be filled otherwise. People may move from areas with a high supply of in-demand skills; however, these moves are costly, and empirically, internal migration rates are not large enough to fill the vacancies created by large increases in specific industries. These jobs may instead be filled by locals with a different (sub-optimal) skill mix, reducing productivity of the industry. An important avenue for future research is understanding how these skill gaps are bridged.

This paper contributes to understanding in several areas of research. First, it provides an empirical basis for the nascent literature endogenizing education choice in models of spatial inequality and mobility. Prior to Eckert and Kleineberg (2021), which to my knowledge is the first to include this adjustment margin in their model, this literature took skill supply as a fixed local characteristic. In this paper, I exploit local changes to establish a supply elasticity for skills in a context where the supply of education does not vary in response to local income changes.

Secondly, this paper relates to an emerging literature understanding how students' choices respond to local labour market conditions. Most closely related is contemporaneous work examining the impact of STEM jobs on high school and college course choice and major in the US (Mather and Smith, 2024). In contrast to this paper, I focus on isolating the student

response in a setting where local conditions do not directly impact secondary education provision. Other papers in this area (e.g. Conzelmann et al. (2023) and Weinstein (2022)) focus on college major choice, using university location to determine the relevant labour market. I consider instead students' earlier location and include their decision whether to attend university in my outcomes.

This work also expands on prior work examining the impact of specific local shocks on educational attainment in various contexts. The existing literature finds that education is a substitute for labour in response to single-sector shocks caused by natural resource booms (Emery et al. (2012), Maurer (2019), and Kovalenko (2023)) and trade policy (Atkin, 2016). I differ from this literature by considering the changes in local composition coming from all sectors, rather than a single industry or policy, and by considering more nuanced educational choices in addition to continued enrolment. My results are also consistent with prior work finding a (permanent) dip in the attainment of those who graduate into a housing-induced local labour market boom (Betts and McFarland (1995), Charles et al. (2018)).

Also related is the literature on the elasticity of education choice to expected earnings. This literature has focused primarily on college major choice in the US, finding a small but significant elasticity³. However, this literature does not explicitly consider the cost of relocation or the spatial distribution of jobs which is the focus of my work.

The findings here also relate to recent work on place-based policies, the hazard cost of which depends on the spatial concentration and willingness to move of the intended transfer recipients (Gaubert et al., 2021). My research points to an additional margin —education choice— in which people in a given location may be entrenched and provides evidence that industrial policy may not change these outcomes for locals.

My results also bring results from agglomeration literature into a new light. Recent work has highlighted the importance of spillovers from significant local industrial investment on the local and total cost of investment policies (Kline and Moretti, 2014). This work finds the largest (positive) spillovers from investment in areas which share worker flows with the industry being funded (i.e. already have a pool of qualified workers) (Greenstone et al., 2010). The muted response I find from local students sheds light on why having an existing

³Altonji et al. (2016) provides a good review of the findings.

local pool of qualified workers is important.

The rest of this paper is organised as follows. Section 2 describes the institutional context of English education. Section 3 describes the datasets and measures used in the analysis, and presents descriptive statistics. Section 4 describes the empirical strategy and assumptions required for the identification strategy. Section 5 presents the findings for attainment and field choice. Section 6 describes additional specifications and extensions. Section 7 concludes and highlights potential avenues for future research.

2 Institutional context

In the UK, education is specialised, and qualifications are standardized at a young age. Education is regulated at the national level; that is, each of England, Scotland, Wales and Northern Ireland have their own requirements. In this paper, I study students in England.

Figure 1 shows the typical progression of students through the English education system. The first opportunity for subject choice comes in Key Stage 4, when students are ages 14-16. At this phase of their education, students choose which subjects to study for their General Certificate of Secondary Education (GCSE) examinations. These exams, typically taken at the end of Year 11 (ages 15-16), are nationally standardised and offered in a variety of subjects. All students are required to take exams in English, mathematics, and science, and schools are required to provide at least one offering in each of arts, humanities, business, and language. Thus, these choices are comparable for all students in England.

Schooling in England is compulsory until age 16. Education is compulsory to age 18; this was increased from 16 to 17 in 2013, and from 17 to 18 in 2015. Post-16 qualifications are standardised across the UK and regulated through the Regulated Qualifications Framework which covers both academic and vocational qualifications. In England, the relevant framework is the National Qualification Framework (NQF). During this period, students have several alternatives for education: academic-track A-levels; technical education; vocational training; on-the-job training; or a mix of volunteering and part-time employment.

In this paper, I focus on whether students continue on the academic track. In contrast to vocational programmes, such as apprenticeships, which require local employment in the

field to be viable, the supply of academic qualifications is less dependent on local demand. Academic disparities, including A-level results and university share, have also been the focus of recent public and political attention.

Following the period of compulsory education, students can choose to continue their education either academically through university courses (which generally require completion of A-levels) or through continuing vocational education. Like academic qualifications, vocational qualifications are regulated and standardized, and thus comparable across time, level, and field.

Students can select up to 5 subjects to study in A-levels but typically take 3 exams, the number required for most university courses. University departments post descriptions of the typical profile of subjects and grades required for admission, which informs students' choices about which subjects to take and where to apply. In practice, this means that the pathways students can take can be restricted by their choice of A-level subjects⁴.

In England, school funding is determined on a per-pupil basis according to the National Funding Formula (NFF); this stands in sharp contrast to the US, where school funding comes primarily from the local tax base. While this formula is adjusted for local economic conditions⁵, it is unrelated to local tax revenue. The funding is distributed to the Local Authority, which then allocates it to schools according to its own formula.

This policy results in school funding on a per-pupil basis being more equitable across schools and less responsive to local conditions than in other contexts. In England, during this time, the 90/10 ratio of per-pupil funding at the school level ranged from 1.3 to 1.6, with the most deprived local authorities having the highest level of funding (Belfield and Sibieta, 2016). This stands in sharp contrast to the US where differences are much larger⁶ and positively correlated with local economic conditions, shutting down an important channel of

⁴Although schools are not required to offer a specific selection of A-levels, students generally have access to a broad variety of subjects. The subjects typically required for most university programmes are offered broadly throughout the country. Figures A2 through A8 show A-level subjects which commonly appear in university requirements. Each map shows the student-weighted share of schools in each TTWA which offer the degree (TTWA with fewer than 3 A-level providers are omitted).

⁵The funding level is adjusted for local (teacher) hiring costs, a measure of income deprivation in students, and the level of remoteness from other services.

⁶For comparison, the 90/10 ratio of per pupil funding in the US for elementary and secondary schools at the more aggregated school district level was 2.4 in 2016 (U.S. Census Bureau, 2021).

inter-generational human capital transmission. Conceptually, this allows me to measure the change in student attainment holding the local education supply fixed.

3 Data and descriptive statistics

3.1 Student data

Student information comes from administrative data from the Department for Education (DfE). Data on qualifications attained during compulsory education, as well as demographic and location information, comes from the National Pupil Database (NPD) (Department For Education, 2023). This is an administrative panel dataset on the census of students attending state-funded secondary schools in England beginning in the 2001/02 school year. Data on university attainment comes from the Higher Education (HE) dataset, which can be linked to records from the NPD. This student data is collected for all students studying a qualification above level 3 (A-Levels or equivalent vocational qualification) at a reporting HE provider.

I attribute to each student the local labour market conditions in the TTWA (PD) in which they lived when they began their GCSE studies (age 14-15, the beginning of Key Stage 4). Throughout, I will refer to this as the TTWA (or PD) which the student is from⁷.

Figure 2a shows the average share of students from each TTWA who enrol in A-levels. The share of students enrolling in A-levels ranges from less than 50% in some coastal areas to nearly 70% in parts of the southeast. This pattern is mirrored in Figure 2b, showing university enrolment, with the share enrolled varying from less than 25% to nearly 50% along the same geographic lines.

In addition to enrolment and attainment, these datasets provide specific information about each subject taken (A-levels) and degree program (university). I map each of these to the fields coded in the Labour Force Survey (LFS). The mapping process is described in more detail in section C.1. Figure 3 shows the field of university enrolment that is most disproportionately popular in each TTWA; that is, the field for which the share of college

⁷Student mobility during this phase of their schooling is generally low. In each cohort, more than 98% of students observed in year 12 live in the same TTWA they did in year 10.

enrolment is most above the national average. This map shows the geographic clustering of different fields, with Business degrees being overrepresented in London and the surrounding areas and broadly-applicable Arts and Humanities degrees overrepresented throughout the southeast.

3.2 Business data

Data on local establishments comes from the Business Structure Database (BSD) (Office for National Statistics, 2024), which is an administrative dataset covering all businesses liable for VAT and/or with at least one member of staff registered for the PAYE tax collection system. This dataset contains approximately 2 million observations annually, covering an estimated 99% of UK economic activity. I observe the location, industry, and employment count at each establishment, as well as the firm to which it belongs.

Although this is an administrative dataset, there are still missing values for many entities. Because of the identification strategy employed, these have the potential to create multiple false 'events' in a given location. To alleviate this problem, I linearly interpolate these values using the surrounding years. This is described in more detail in the data appendix.

3.3 Other data

To link the employment from the BSD to the skills mix in the DfE data, I use the existing skill composition of young workers (ages 25-34) in each industry from the Labour Force Survey (LFS) (Office For National Statistics, 2023). The LFS is a nationally representative survey designed to provide information on the UK labour market. In addition to employment information, including detailed industry, it includes detailed information on qualifications held by individuals. I use information on the level and field of the highest level of education. The sample of younger workers is used to more closely approximate the skills requirements for the current students, and for increased consistency in qualification standards over time.

3.4 Data linking

Because the BSD includes only headcount by employer, not information about employees, I must impute the local qualification distribution using data from the LFS. I describe this process in more detail in section 4.1. Linking the BSD to the LFS by industry allows me to translate the industry employment information in the BSD into estimated employment by qualification.

I aggregate the student data by geography, calculating the total number of students who take each type of qualification discussed above. I then match this geography x time panel to the business data to create a novel dataset linking student outcomes and business demand.

4 Empirical strategy

To estimate the student response to changes in local labour market conditions, I consider students living in location (local labour market) l at time t , measure changes skill s in location l , and observe the students' academic outcomes in later years.

4.1 Measuring local skill demand

The first step in measuring local skill demand is determining the qualifications used in each industry. I compute this using the empirical distribution of held by workers ages 25-34 in industry j in the LFS. Specifically, I sum the weighted person count by qualification type (field or level) for each industry and divide by the total weighted person count in the industry. This creates a skill share vector, $\mathbf{skills}_j$, for each industry j .

I also compute the local industry employment shares, $emp_{j,l,t}$, using BSD data. This is simply the total employment in industry j and location l at time t .

I then combine these elements to aggregate these industry skill vectors and calculate the skill demand in location l at time t . This is calculated as the employment-weighted sum of skills vectors in the industries in that location and time.

$$\mathbf{skills}_{l,t} = \sum_{j \in J} emp_{j,l,t} * \mathbf{skills}_j \quad (1)$$

Here, $\mathbf{skills}_{l,t}$ is the vector of skills used in employment in location l at time t . $emp_{j,l,t}$ is the employment in industry j in l at time t . $\mathbf{skills}_j$ is industry skill vector discussed above. The assumption embedded in this measurement is that the skill requirements for new jobs in the industry are the same as those held by existing young workers.

The industry-specific skills vectors are derived from LFS data as described in Section 3.3. Thus, the actual local skill composition is estimated using the national industry-average skill mix. While this approach was a necessity due to data availability, it has the advantage that it abstracts one level from skill supply decisions in firm moves. That is, the local area need not have an existing supply of workers with a specific skill for that skill to be recognised as in-demand by the industry. This approach can be thought of as analogous to a shift-share instrument, where there shares are the industry skill mix and the shifters are the local industry composition. This interpretation is discussed in more detail in section 5.2.

For tractability, I consider separately the level and field of skills demanded. For outcomes relating to student enrolment decisions and attainment, I consider only the level of qualifications required. For these specifications, the skills vector is $[Degree, Non - degree]$. For specifications relating to field choice, I use the field composition of qualifications⁸ seen in Figure 4b.

Figure 4a shows the average expected share of employment in a TTWA requiring a degree. As with the student data, rates of university attendance are much higher in the South East. Figure 4b shows the field of study that is most disproportionately represented in the industry mix in each TTWA. As with the student data, Business degrees are overrepresented in London. Areas with lower degree employment also see a disproportionately high share of activities typically associated with rural areas, as well as services (health and education) that are provided locally.

4.1.1 Average skill share

In my correlational results, I consider the average skill share in the three years prior to a decision. I take this measure to smooth any transient changes in labour market shares and

⁸The LFS attributes field only to qualifications level 3 (A-Levels or equivalent vocational qualification) and above in the NQF, thus, the field mix I use is that of qualifications at this level or higher.

because the exact timing of the relevant labour market characteristics is ambiguous.

4.1.2 Changes in the overall local skill composition

I utilize the change in local skill mix coming from changing local industrial composition. The change I compute is the one-year difference in skill share. $skills_{s,l,t}$ is the component of the skills vector $skills_{l,t}$ in skill s , and $||skills_{l,t}||$ is size of the workforce in location l at time t . Thus, $\Delta skills_{s,l,t}$ is the one year change in the percentage of workers with a given skill.

$$\Delta skills_{s,l,t} = \frac{skills_{s,l,t}}{||skills_{l,t}||} - \frac{skills_{s,l,t-1}}{||skills_{l,t-1}||}$$

To better understand the dynamics of changes in skill composition, I construct an indicator based on the change in skills in the local area. This allows me to compare cohorts who are yet to make a decision with those already past the start of the qualification.

$$D_{s,l,t} \equiv \mathbb{1}(\Delta skills_{s,l,t} > \Theta)$$

Change threshold	TTWA	PD
Θ		
Degree share	0.010	0.015
Field share	0.050	0.075

I chose a value of Θ to maximize the number of locations which experienced exactly one shock. The results and interpretation are not sensitive to alternative threshold choices.

To test the validity and observe the skill share dynamics of this change, I consider the following specification:

$$skills_{s,l,t} = \sum_{\tau=-4}^6 \left(\beta_{\tau}^{emp} D_{s,l,t+\tau} \right) + \gamma_{s,t} + \alpha_{s,l} + \epsilon_{s,l,t}$$

Figure 5 shows the dynamics of degree share employment around a large increase in the degree share. Figure 6 shows the same for fields of study. While, in both cases, the share in the years preceding the increase is significantly above that in the year immediately prior,

the years after the increase are significantly higher than the earlier level. The overall change shows only a modest attenuation in the 5 subsequent years and remains well above the pre-increase levels. This indicates a sustained change in the share of employment requiring the specified skill.

4.2 Student response

First, for the correlational estimates, I consider how students' choices relate to the level of employment in a skill in their TTWA in the 3 years preceding the decision point, $skills_{s',l,t+\tau}$.

$$Sh_{s,l,t} = \beta \sum_{\tau=-3}^{-1} \left(skills_{s',l,t+\tau} \right) / 3 + \gamma_{s,t} + \epsilon_{s,l,t} \quad (2)$$

$Sh_{s,l,t}$ is the share of student cohort in location l at time t who choose to study for qualification s . τ is 0 when the student is in the first year of schooling after the decision is made. For example, for the decision to attend university, $\tau = 0$ in a student's last year of A-levels. Take-up of a skill may change in response to demand changes for s , or for related skills s' ; for example, A-level enrolment (s) is a prerequisite for degree enrolment, so an increase in jobs requiring degrees (s') can be used on the right-hand side. In all specifications, the data is at the Location x Skill x Year level. Observations are weighted by the total number of students in the TTWA in the initial year (2002). Standard errors are clustered at the TTWA level to account for the correlation of field shares within a location at a given time, and the correlation across time within a location from the averaging of local employment shares.

For the causal estimates, I estimate the following Difference-in-Differences specification:

$$Sh_{s,l,t} = \sum_{\tau=-3}^4 \left(\beta_{\tau}^{incr} D_{s',l,t+\tau} \right) + \gamma_{s,t} + \alpha_{s,l} + \epsilon_{s,l,t} \quad (3)$$

Here, the right-hand side variables of interest are lags and leads of the dummy for large increases described above. I also include Skill x Time and TTWA x Skill fixed effects. The unit of observation, weighting, and standard errors are the same as in the correlational estimates; here it is necessary to cluster across time within a location due to the local skill share fixed effect ($\alpha_{s,l}$).

5 Results

In this section, I first show the relationship between local skill demand and student education outcomes. I then discuss the assumptions required for the causal estimates that follow and present students' responses to changes in their local labour market.

5.1 Correlational evidence

5.1.1 Attainment and local degree share

First, I consider the relationship between local degree employment and student attainment. Table 1 shows the correlations at the TTWA level in both levels and differences⁹. TTWAs with a higher share of degree employment have a significantly higher share of students enrolling in A-levels, more A-level passes (per student from that area), and a higher share of students enrolling in university. Compared to TTWAs in the 25th percentile, those in the 75th have 10pp more students enrolling in A-levels and 8pp more students enrolling in university. Looking at the 5-year change in local degree employment, the sign of the coefficients remains positive, but the magnitude is much smaller and they are no longer significant. Most of the differences come from variations between, rather than within, TTWAs.

5.1.2 Field choice and local field share

I next consider the relationship between local field employment and students' field choice. Table 2 shows the correlations at the TTWA level in both levels and differences. TTWAs with a higher share of field employment have a significantly higher share of students enrolling in that field in university; the relationship does not hold for A-levels, which are much broader subjects. For every 10pp higher the field share of an area is compared to others, the share of students from that area studying in that field is 3pp higher. Looking at the 5-year change in local field employment, the sign of the coefficients reverses, indicating that students may substitute education for employment within a field.

⁹Table A1, shows how these margins respond to local labour market conditions as a whole.

5.2 Inference and estimation challenges

To understand the dynamics of the changing labour demand, I use the estimators developed in De Chaisemartin and D’haultfoeulle (2023). For these estimates to be interpreted as causal, there must be no anticipation effects, and the parallel trends assumption must hold.

In my specifications, $t = 0$ is the cohort that had just made a decision about the outcome variable, and thus, in the absence of anticipation effects, is unaffected. I report coefficients for the three preceding cohorts (who were already past the decision threshold at the time of treatment) as well and see estimates indistinguishable from 0 in all cases. To the extent the immediately preceding cohort anticipates the employment change (perhaps due to newspaper articles, etc.), the earlier cohorts would have to have the same anticipation, which seems unlikely given the nature of the changes.

In my specifications, I allow for nonparametric differences in the evolution of different fields over time. The parallel trends assumption is therefore within skill (e.g. the expected evolution of Natural Science degrees is the same for treated and untreated groups; it need not be the same as the evolution of Arts degrees). One potential cause for violation of the parallel trends assumption would occur if firm activity driving the events was a response to changing local circumstances. For example, if firms saw an increasing share of students studying for relevant skills, and expanded as a result; triggering an event in areas that were different from untreated areas. However, this would require firms to have detailed knowledge of student choices at a local level, and an assumption that this increase would persist. I also do not observe any evidence for this scenario in the pre-event periods.

The variation I exploit can also be interpreted as a shift-share instrument for local skill demand, where the shares are the national skill vectors in each industry and the shifter is the change in local industrial composition. This interpretation helps to alleviate concerns about the endogeneity of increases in a specific skill, as the measured change is mediated by the industrial skill composition. However, by using this approach, actual changes in local skill demand are not fully captured. If, for example, the national (or regional) skills used in an industry are not representative of the skills required to be held by workers, the variation I use would not be relevant to students preparing to enter the labour force.

The other plausible shift-share instrument for this work is to shift the local industry composition (and thus skill composition) using the change in national industry composition (i.e. assuming any national shifts are proportional across space). This would address endogeneity concerns about the growth of industries in specific locations as opposed to others. However, it has the drawback (particularly for contractions in local demand for a skill) that it entangles local and national demand.

5.3 Causal evidence

I find muted responses to local labour market conditions, and responses to hyper-local (post-code district) conditions different than those at the local labour market (TTWA) level.

5.3.1 Attainment and local degree share

I first consider how enrolment and attainment respond to an increase in the share of local jobs requiring degrees. A-levels are the most common path to university enrolment; therefore, if the choice to enrol in A-levels was primarily driven by forward-looking skill concerns, enrolment would grow in response to increased demand for jobs requiring degrees.

Figure 7 shows how enrolment in A-levels changes in response to an increase in local degree employment, as specified in equation 3. I find no evidence that students' choice to enrol in A-levels responds to an increase local degree demand.

I then turn to the composition in students' more immediate vicinity. Figure 8, shows the response at the PD level. The positive point estimates increase a change of around 0.35pp in the first two years and 0.90pp in the following two. The confidence intervals rule out an increase of more than 2.5pp, equivalent to one student per cohort for every 10 new jobs. With an average increase in local employment of 2.5pp in the share of jobs requiring a degree, this rules out an increase even at replacement level for the new skill composition.

The next margin of choice students face is whether to attend university. Figure 9 shows how the share of students who enrol in a degree program changes in response to increase in local degree employment. Again, I find no evidence that enrolment changes following an increase in degree employment ¹⁰.

¹⁰Figure A1, shows how these margins respond to local labour market conditions as a whole. I find no

5.3.2 Field choice and local field share

Finally, I consider the field choice of university students. Figure 10 shows the share of students who study that subject in university. Here, I also find no evidence that, in the short term, students respond to changing local demand. The estimates also bound the change at no more than an 0.4pp increase for every 1pp increase in field share. At the PD level, there is a borderline-significant increase in field enrolment, as seen in figure 11. However, the magnitude is small: it is bounded from above by a 0.25pp increase in enrolment for every 1pp increase in field share in jobs. This amounts to 1 new student in a field for every 200 new jobs.

All together, my results show that any short-run student response to changing local conditions is muted, and that the change in sills from local students is insufficient to meet the new demand.

5.4 Heterogeneity

I now consider how these results differ for different types of students. First, I consider the type of school where the students attended A-levels. In England, different types of schools have different levels of autonomy. Here I consider separately students enrolled in community schools, which are funded and fully administered by the local authority, and those enrolled in academies (including converters) which are funded by the DfE but independent of the Local Authority. Figure 12 shows the response limiting the sample to students who, in Year 12, were enrolled in an academy. Here, I find noisy estimates and cannot rule out no response. However, the magnitude of the point estimates is much larger than for the total sample, and warrants further study. Figure 13 shows the results for only students enrolled in community schools. As in the full sample, the magnitude of the response is small enough to be indistinguishable from 0 - I find no evidence that students in these schools respond to changes in local demand.

I then consider whether these results differ by the type of university attended. This is an endogenous choice for students, so the causal interpretation is less clear. I consider the evidence that students substitute away from university enrolment when the local economy expands.

subject choices of students who attend Russell Group universities and those who attend former polytechnics. Those who attend Russell Group universities are much more likely to live outside their home TTWA after university (Britton et al., 2021), and therefore may be expected to respond less to local demand. Figure 14 shows the field choice response for students enrolled in Russell Group universities, and 15 shows the same for those enrolled in former polytechnics. In both cases, there is no evidence for an increase in field enrolment following an expansion of the local field share.

6 Additional specifications

6.1 Different skills

My current analysis focuses on academic qualifications — A-levels and university — but there is also a unified qualification framework for vocational and technical skills. Students who do not attend university are more likely to remain in the area where they grow up, so the local conditions may be more relevant for them. These skills are relevant for policies such as Reshore UK, intended to relocate manufacturing jobs from abroad to the UK. The response to demand for these skills may be different due to different training modes, and may also be more dependent on local training supply. The endogeneity of the availability of these programmes will require careful empirical analysis.

6.2 Student characteristics

In addition to using different variations in the skills demanded, it is also relevant to examine how different types of students respond to changes in local demand.

6.2.1 Earlier performance

Graduates from the most selective group of universities¹¹ are more mobile than graduates of other universities, with those from the least selective universities being the least likely to

¹¹The most selective universities are defined as the Russell Group, a self-selected group of twenty four research universities. These universities receive the bulk of all research funding.

move (Britton et al., 2021). Another fruitful extension could therefore be to investigate how these responses vary with students' earlier performance by splitting students into groups based on their results before the decision to enter A-levels.

6.2.2 Geographic mobility

In the current analysis, I consider all students that are living in a place at a given time (before the local economic changes). I can also restrict the analysis to students who remain in that area. Conversely, I could identify the area effects on students who move with a measure of 'exposure' to different subjects à la Chetty et al. (2016).

6.2.3 Demographics

While the primary focus of this analysis is the drivers of inequality across space, the drivers of education choice and responsiveness to local changes may also contribute to inequality within a location. Students who are more responsive may differentially benefit from higher returns to in-demand local skills (Altonji et al. (2012) Early et al. (2020)). Although I do not observe the parents' industry, I will use an indicator for whether the student was eligible for Free School Meals (FSM) as a measure of familial background. This will allow me to identify whether the responsiveness is different by parental background.

6.3 Broader changes

This analysis has focused on local changes, controlling for national circumstances, rather than on broader demand. Particularly for university decisions, the overall prospects may have an important impact. Another avenue for future research is using a shift-share instrument to how national shocks, such as import competition, affect students' choices.

7 Conclusion

In this paper, I study spatial differences in student attainment, achievement, and field choice. I find that students' choices are highly correlated with those used in jobs in their local area,

but, in the short run, do not significantly respond to changes in local demand. This has important consequences for the results of policies implemented with the goal of bringing good jobs to left-behind communities.

Multiple mechanisms could be at play causing these differences: the local changes may not be salient, the types of skills required may not be those that students had considered, or students may be considering other factors in their choice of study. Regarding salience, I find that even the most local of changes (on average within 5 miles of students' location) do not significantly affect take-up. To further address this concern, future research can instead consider large expansions or contractions of prominent local firms, which may be more noticeable to local students.

While there is no mechanical relationship between one cohort and the next there, could also be inertia at the local or school level hampering changes. For example, teachers may be more familiar with the pathways that previous cohorts took. Furthermore, there are other local factors, such as parental composition, that may push students to follow the more traditional local path (Ventura, 2023). Understanding these factors leading to local persistence of education choices is a potentially fruitful area for further research.

While the English setting is ideal for understanding student responses holding constant local education provision, this needs to be considered when applying of this research to other contexts. Expectations of the total student response to changing local circumstances would also need to consider local skill provision, as the response I measure is that of students when local provision does not change. This may be complicated by latent demand for skills which were not previously offered, resulting in in higher take-up of these avenues than I observe in England.

The muted response of local students to changing conditions leaves many questions open for future policy research. If the goal of these policies is for the local population to gain valuable skills, it is important to know whether coordinating job growth policies with targeted upskilling programs may induce a response not seen with changing job skill composition alone. It is also relevant to understand whether other skills, such as those requiring vocational training, which are more likely to be used locally are likely to respond to changes.

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Figure 1: England education timeline

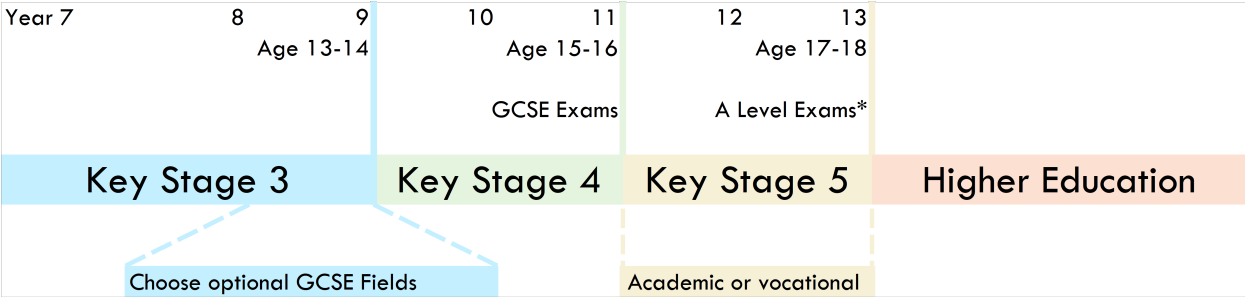
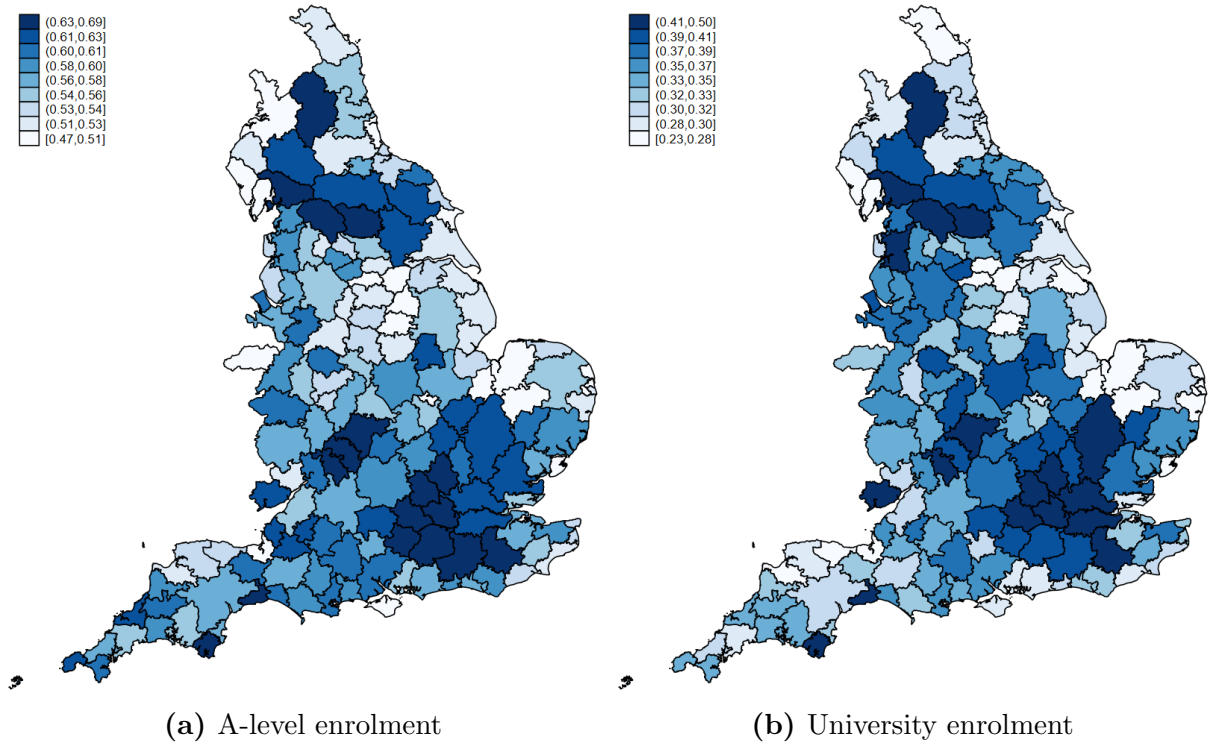
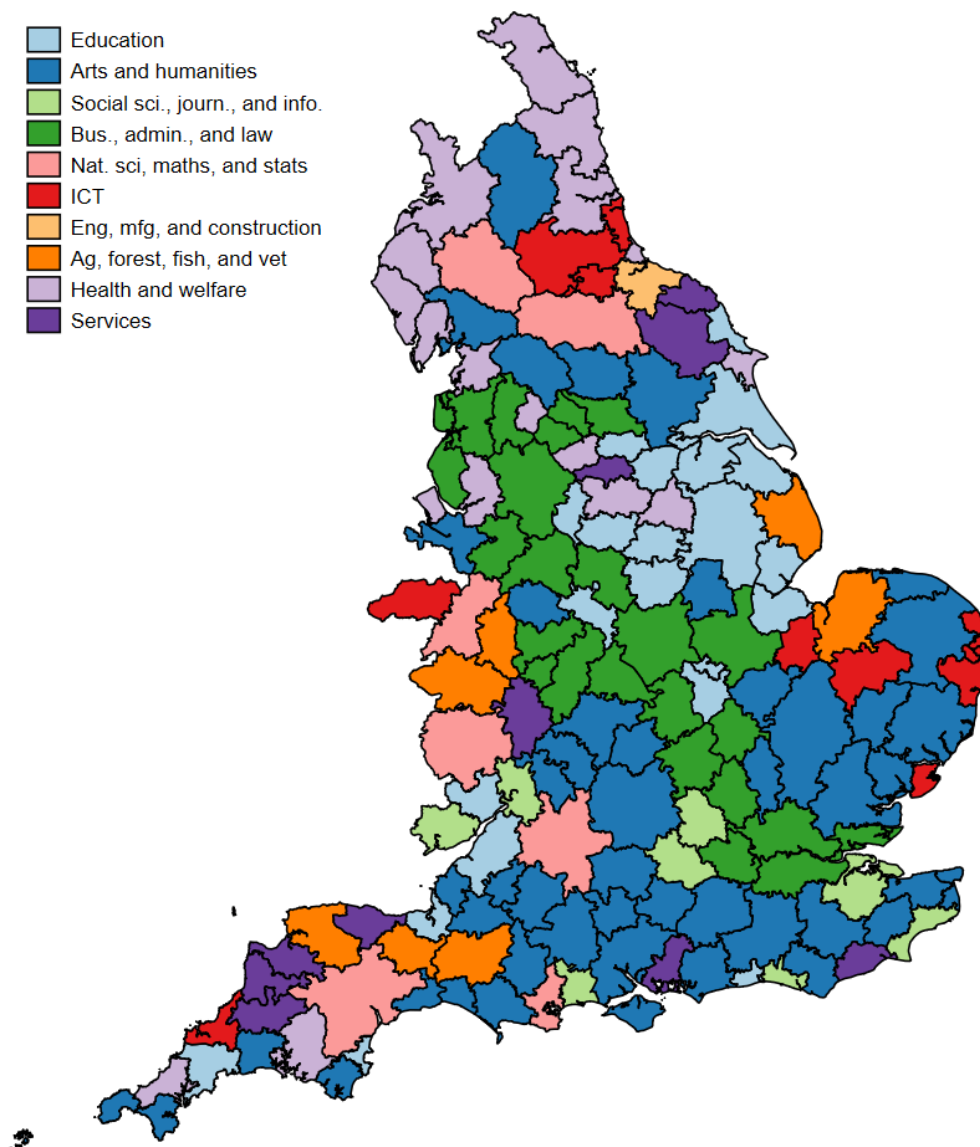


Figure 2: Local student enrolment



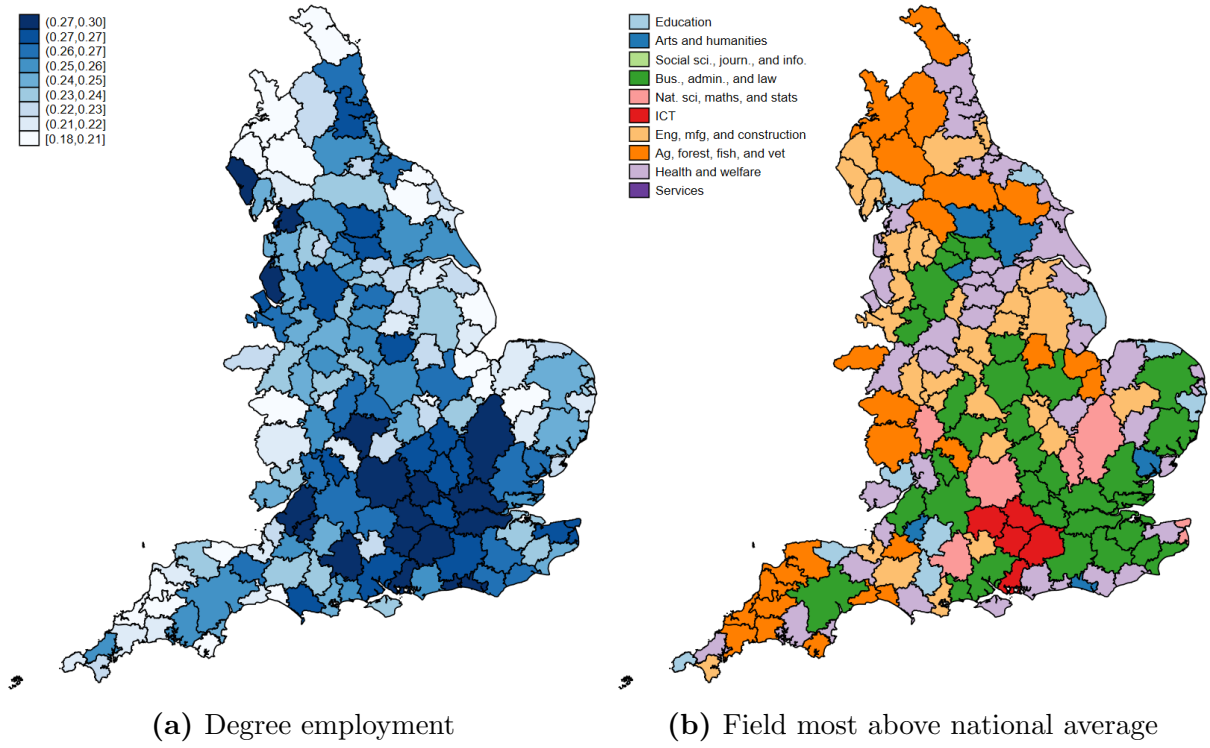
Student information from the NPD. Cohorts that were in Year 10 from 2004-2013 are included. Student information for HE enrolment comes from HESA. For university enrolment, cohorts that were in Year 10 from 2005-2012 are included. Students are allocated to the first degree of the highest level in which they enrol within 9 years of Year 10.

Figure 3: Local students' most prominent university field choice



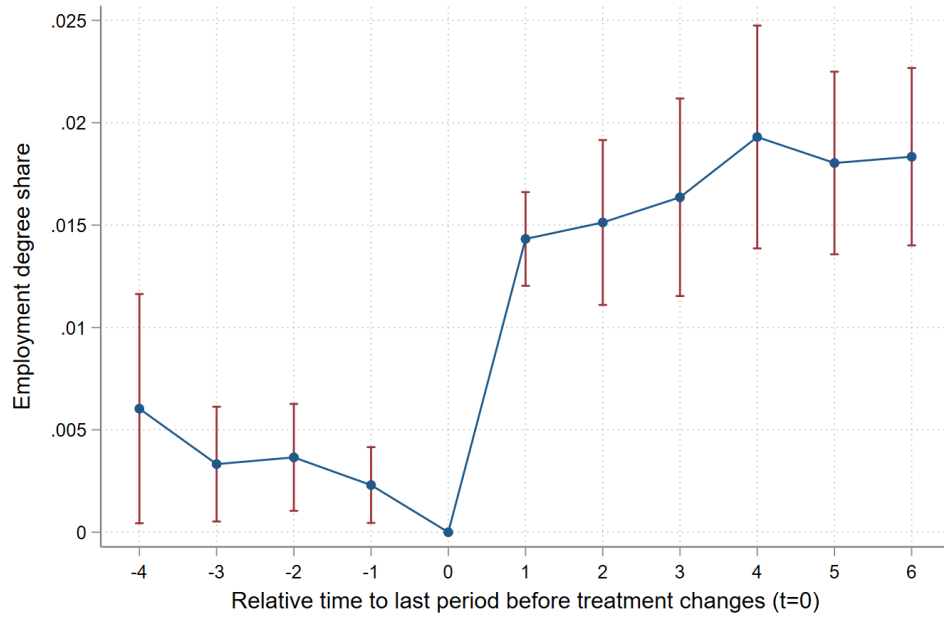
For each TTWA, the map shows the field of study that is most overrepresented compared to national levels. Student information for HE enrolment comes from HESA. For university enrolment, cohorts that were in Year 10 from 2005-2012 are included. Students are allocated to the first degree of the highest level in which they enrol within 9 years of Year 10.

Figure 4: Local employment shares



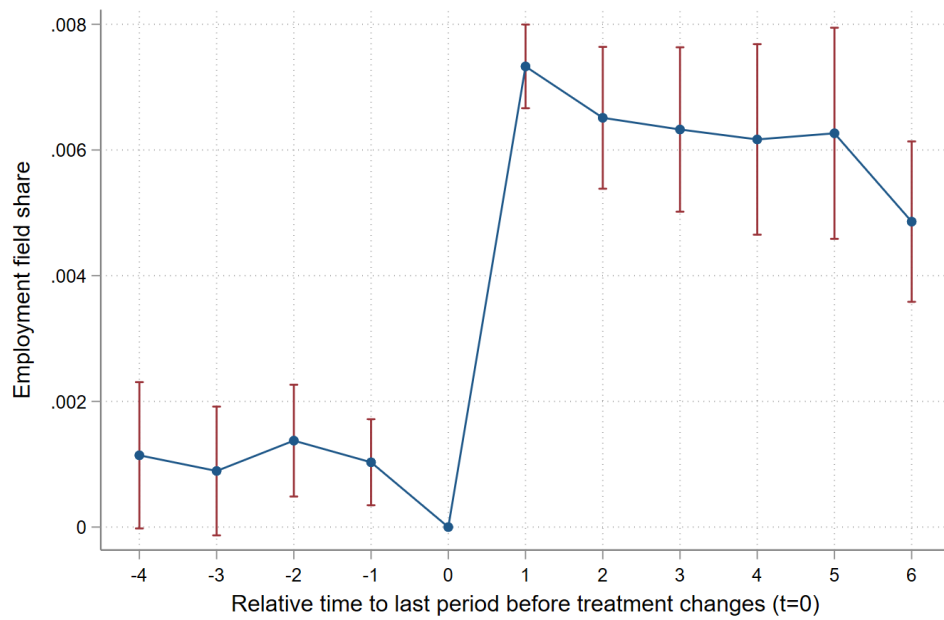
Employment shares from the BSD, combined with the LFS. For each TTWA, the map in panel (b) shows the field of employees in that location that is most overrepresented compared to national levels.

Figure 5: Degree employment shares increase following degree expansion



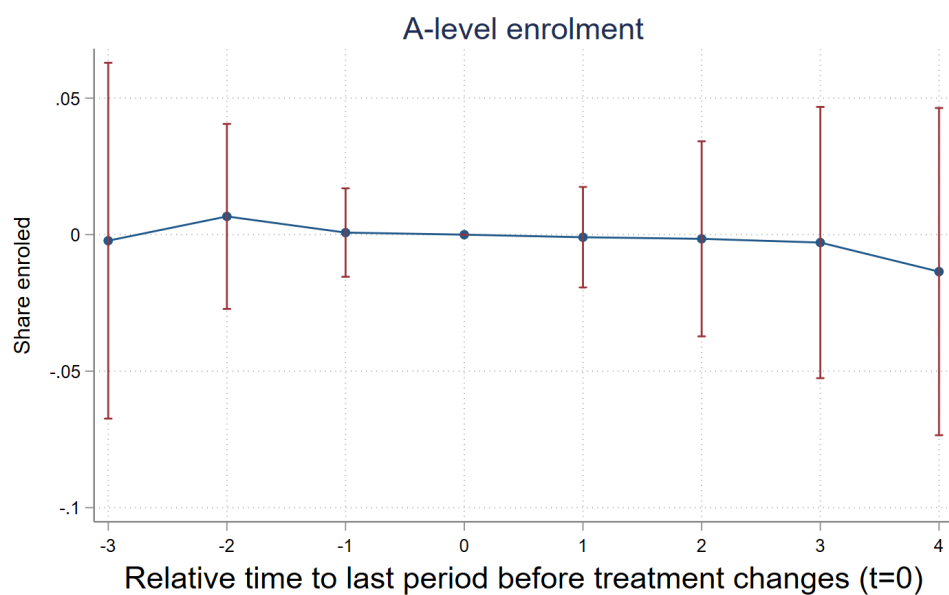
Data at the TTWA x Year level. Employment shares and changes from the BSD.

Figure 6: Field employment shares increase following field expansion



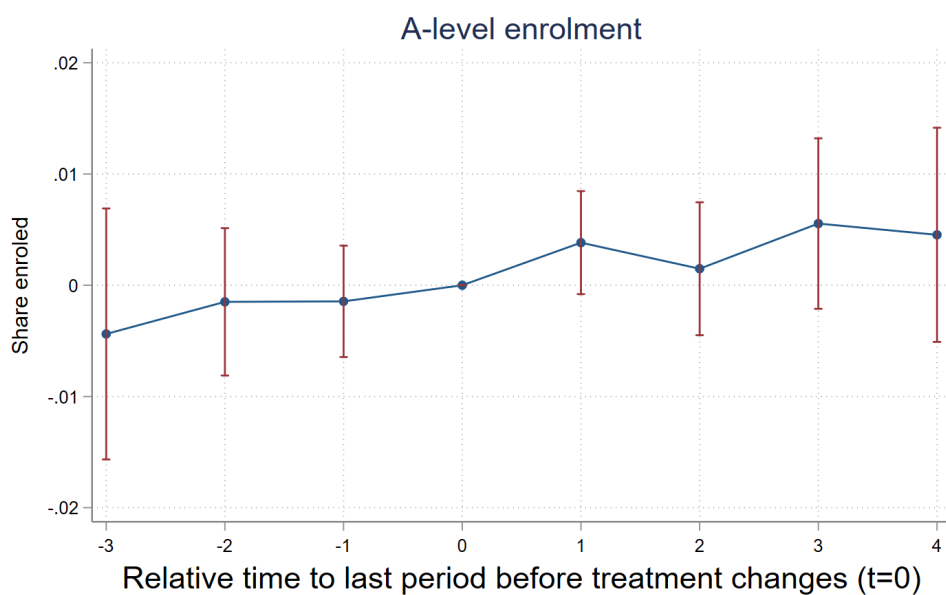
Data at the TTWA x Year x Qualification level. Employment shares and changes from the BSD.

Figure 7: A-level enrolment following an expansion of local (TTWA) degree employment



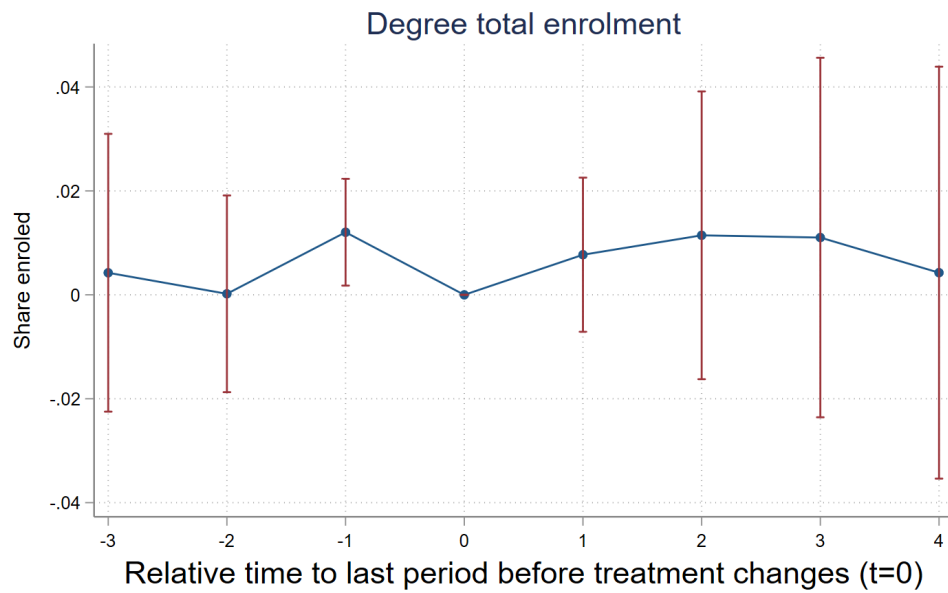
Data at the TTWA x Year level. Employment shares and changes from the BSD. Student information from the NPD. Cohorts that were in Year 10 from 2004-2013 are included. RHS variable used is an indicator for changes of more than 1.0pp in the share of jobs in an area that employ a worker with a degree. Standard errors are two-way clustered by base year TTWA.

Figure 8: A-level enrolment following an expansion of Postcode District degree employment



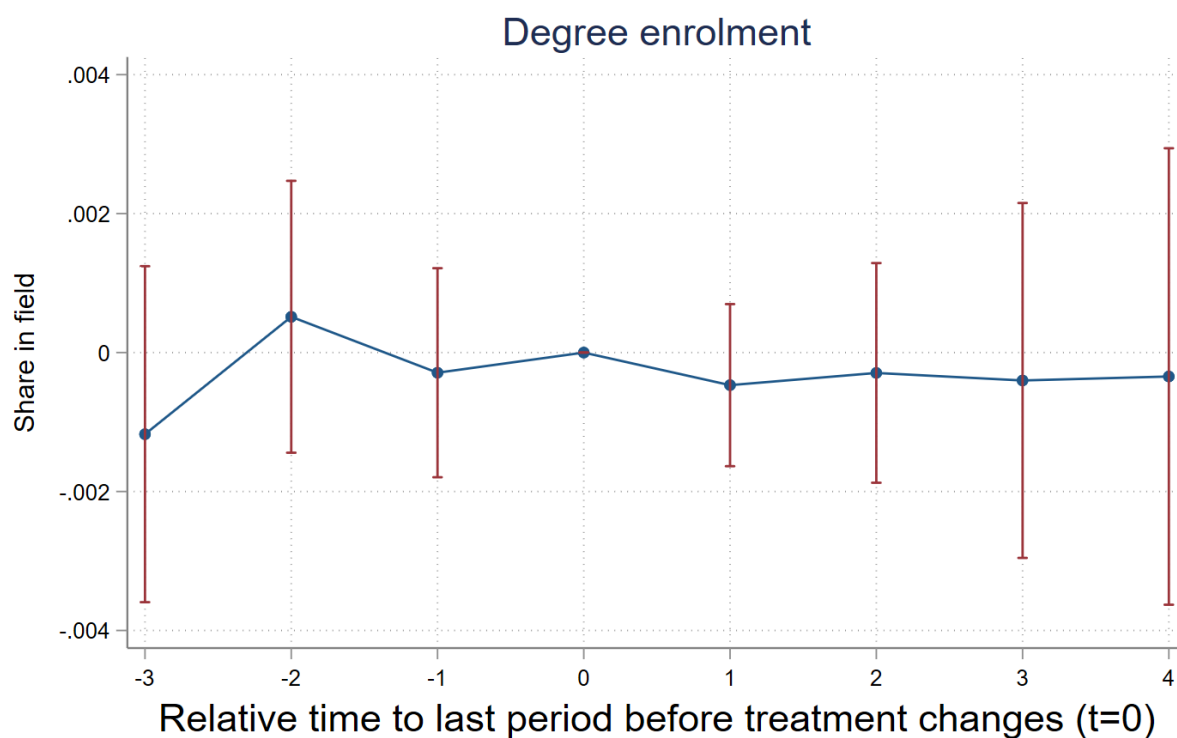
Data at the Postcode District x Year level. Employment shares and changes from the BSD, student shares from the NPD. Student information from the NPD. Cohorts that were in Year 10 from 2004-2013 are included. RHS variable used is an indicator for changes of more than 1.5pp in the share of jobs in an area that employ a worker with a degree. Standard errors are two-way clustered by base year TTWA.

Figure 9: Degree enrolment following an expansion of local degree employment



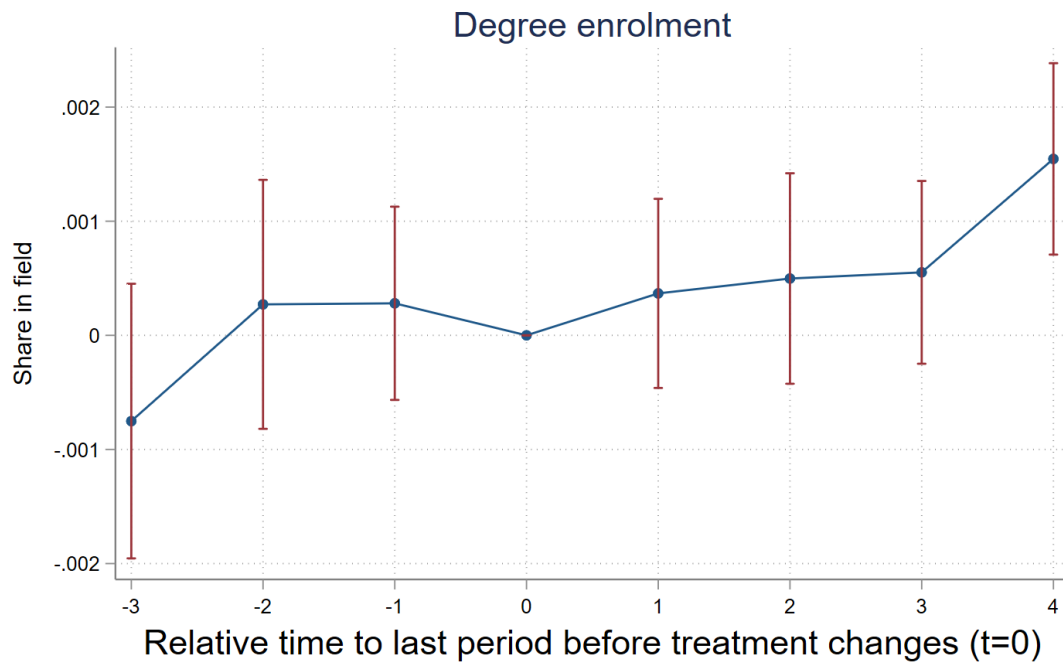
Data at the TTWA x Year level. Employment shares and changes from the BSD. Student information for HE enrolment comes from HESA. For university enrolment, cohorts that were in Year 10 from 2005-2012 are included. Students are allocated to the first degree of the highest level in which they enrol within 9 years of Year 10. RHS variable used is an indicator for changes of more than 1.0pp in the share of jobs in an area that employ a worker with a degree. Standard errors are two-way clustered by base year TTWA.

Figure 10: University field selection following an expansion of local field employment



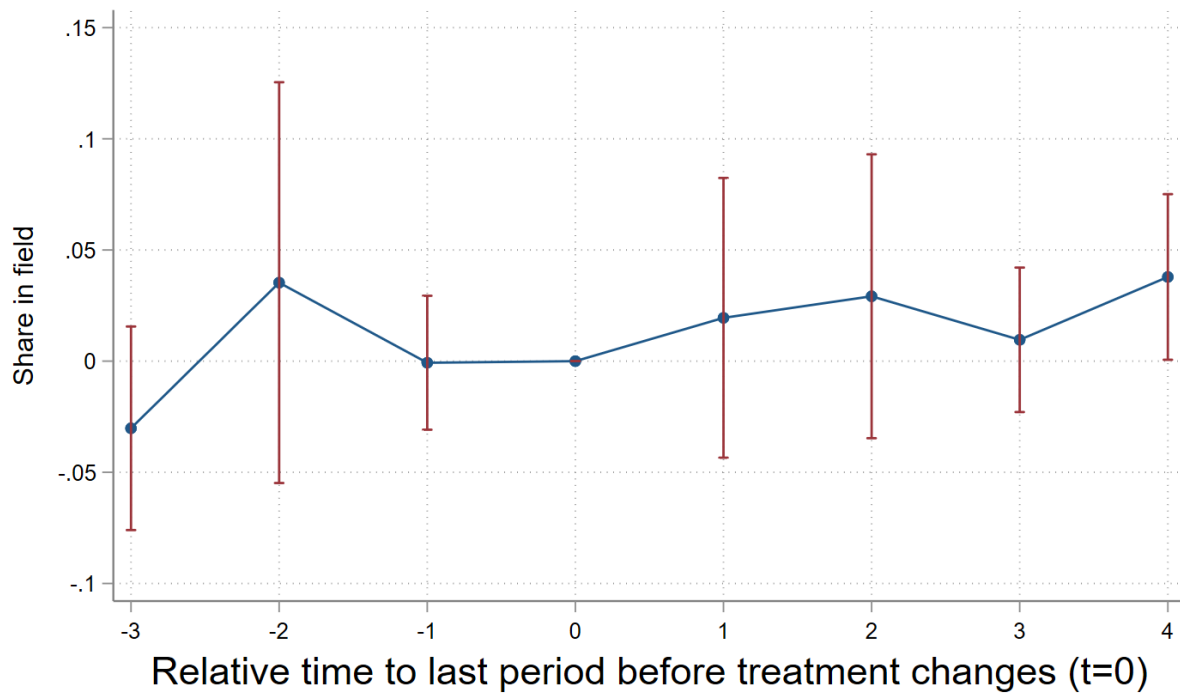
Data at the TTWA x Year x Qualification level. Employment shares and changes from the BSD. Student information for HE enrolment comes from HESA. For university enrolment, cohorts that were in Year 10 from 2005-2012 are included. Students are allocated to the first degree of the highest level in which they enrol within 9 years of Year 10. RHS variable used is an indicator for changes of more than 0.5pp in the share of jobs in an area that employ a worker with a qualification in the field. Standard errors are two-way clustered by base year TTWA.

Figure 11: University field selection following an expansion of Postcode District field employment



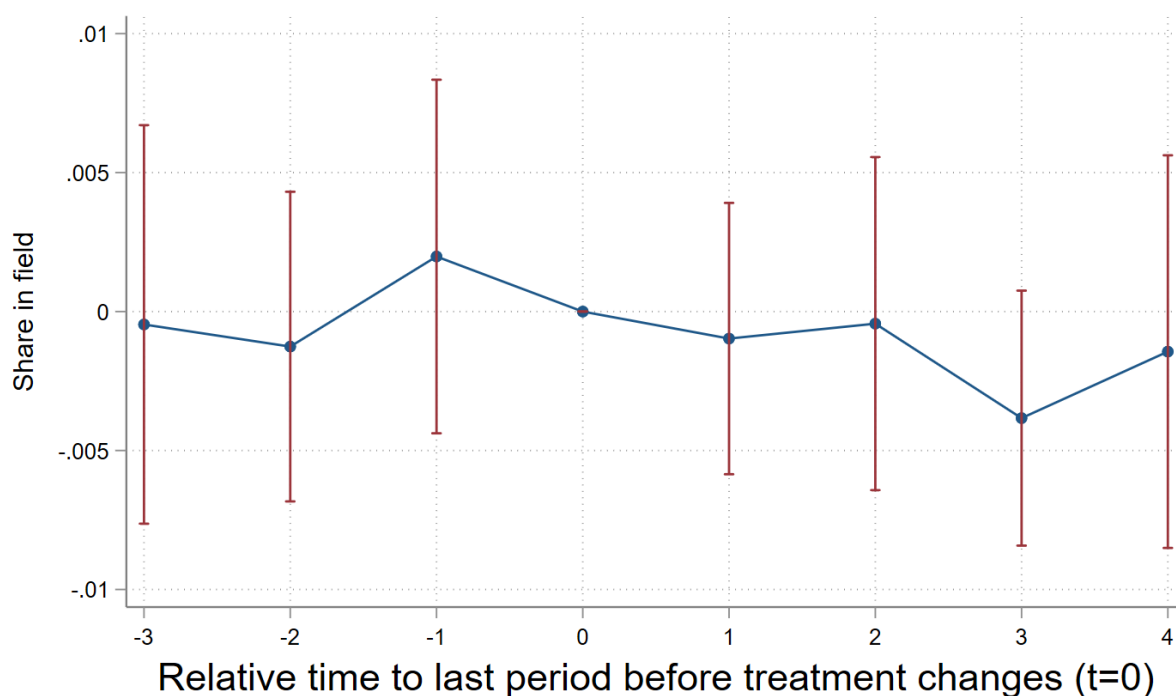
Data at the Postcode District x Year x Qualification level. Employment shares and changes from the BSD, student shares from the NPD. Student information for HE enrolment comes from HESA. For university enrolment, cohorts that were in Year 10 from 2005-2012 are included. Students are allocated to the first degree of the highest level in which they enrol within 9 years of Year 10. RHS variable used is an indicator for changes of more than 0.75pp in the share of jobs in an area that employ a worker with a qualification in the field. Standard errors are two-way clustered by base year TTWA.

Figure 12: University field selection for academy students following an expansion of local field employment



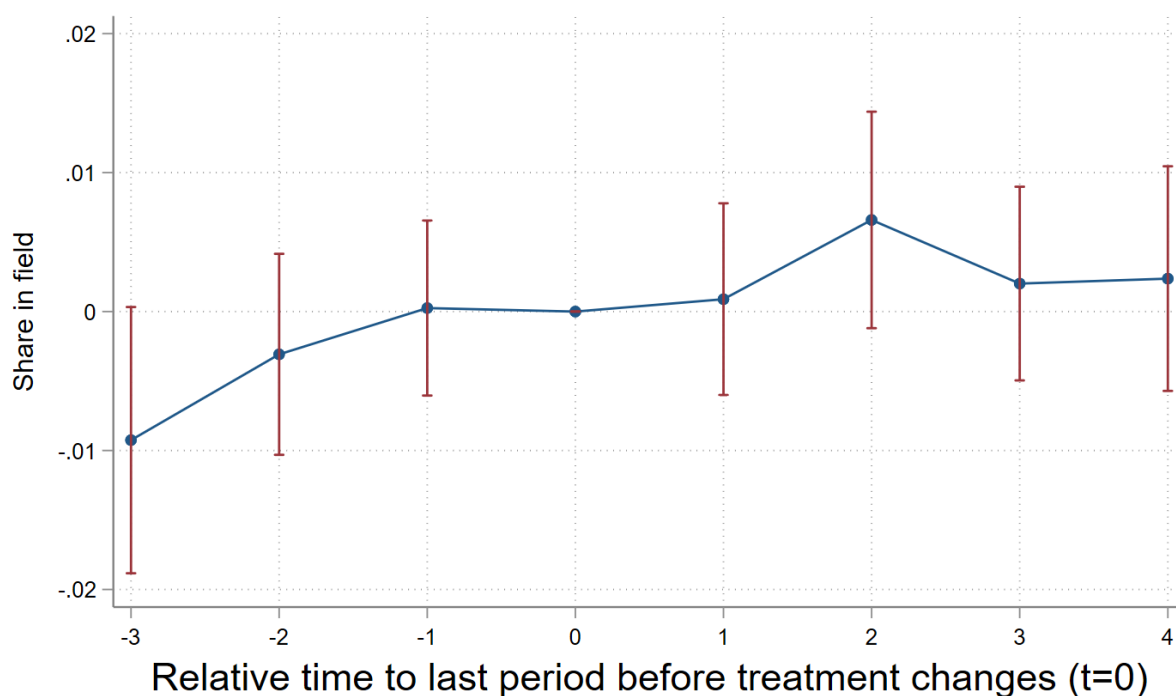
Data at the TTWA x Year x Qualification level. Employment shares and changes from the BSD. Student information for HE enrolment comes from HESA. For university enrolment, cohorts that were in Year 10 from 2005-2012 are included. Students are allocated to the first degree of the highest level in which they enrol within 9 years of Year 10. Only students who attended an academy in Year 12 are included. RHS variable used is an indicator for changes of more than 0.5pp in the share of jobs in an area that employ a worker with a qualification in the field. Standard errors are two-way clustered by base year TTWA.

Figure 13: University field selection for community school students following an expansion of local field employment



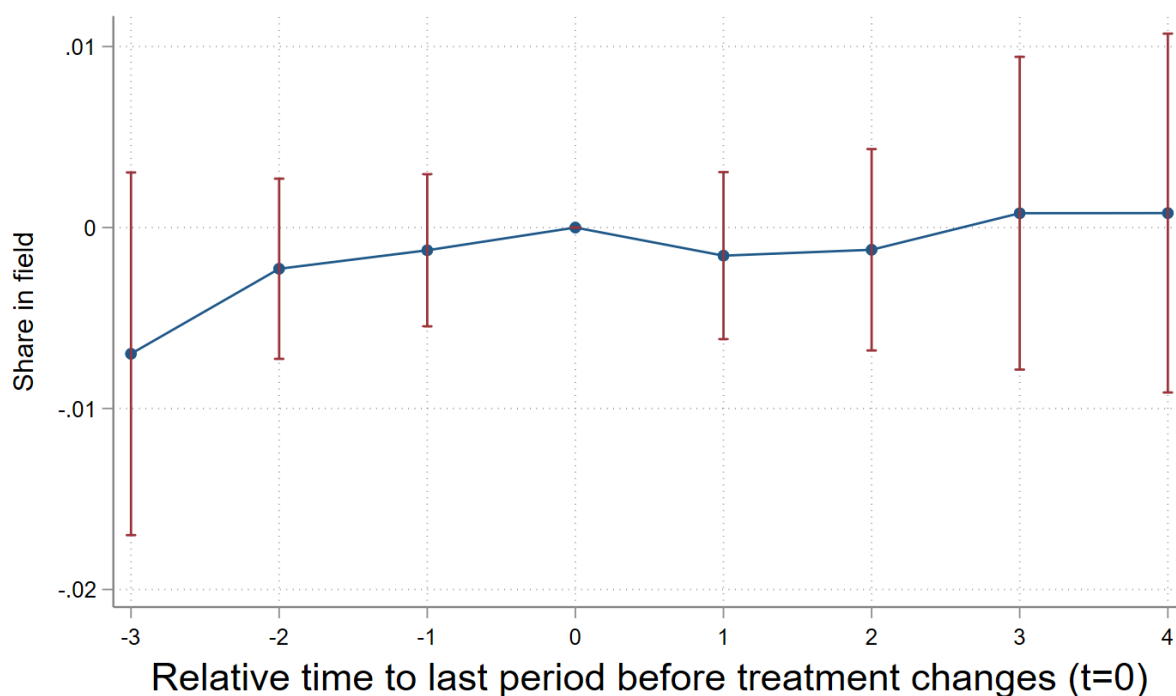
Data at the TTWA x Year x Qualification level. Employment shares and changes from the BSD. Student information for HE enrolment comes from HESA. For university enrolment, cohorts that were in Year 10 from 2005-2012 are included. Students are allocated to the first degree of the highest level in which they enrol within 9 years of Year 10. Only students who attended a community school in Year 12 are included. RHS variable used is an indicator for changes of more than 0.5pp in the share of jobs in an area that employ a worker with a qualification in the field. Standard errors are two-way clustered by base year TTWA.

Figure 14: University field selection for Russell Group students following an expansion of local field employment



Data at the TTWA x Year x Qualification level. Employment shares and changes from the BSD. Student information for HE enrolment comes from HESA. For university enrolment, cohorts that were in Year 10 from 2005-2012 are included. Students are allocated to the first degree of the highest level in which they enrol within 9 years of Year 10. Only students who attended a Russell Group are included. RHS variable used is an indicator for changes of more than 0.5pp in the share of jobs in an area that employ a worker with a qualification in the field. Standard errors are two-way clustered by base year TTWA.

Figure 15: University field selection for students at former polytechnic universities following an expansion of local field employment



Data at the TTWA x Year x Qualification level. Employment shares and changes from the BSD. Student information for HE enrolment comes from HESA. For university enrolment, cohorts that were in Year 10 from 2005-2012 are included. Students are allocated to the first degree of the highest level in which they enrol within 9 years of Year 10. Only students who attended a polytechnic university are included. RHS variable used is an indicator for changes of more than 0.5pp in the share of jobs in an area that employ a worker with a qualification in the field. Standard errors are two-way clustered by base year TTWA.

Table 1: Relationship between local degree share and student enrolment

	A-Level Enrol.	A-Level Passes	Uni Enrol.	A-Level Enrol.	A-Level passes	Uni Enrol.
Share degree emp.	0.544*** (0.118)	1.566** (0.549)	0.409** (0.125)			
Δ_5 Share degree emp.				0.157 (0.154)	0.390 (0.660)	0.0890 (0.157)
Year FE	Y	Y	Y	Y	Y	Y
N	1705	1705	1705	930	930	930

Data at the TTWA x Year level. Employment shares and changes from the BSD. Student information from the NPD. Cohorts that were in Year 10 from 2004-2013 are included. Student information for HE enrolment comes from HESA. For university enrolment, cohorts that were in Year 10 from 2005-2012 are included. Students are allocated to the first degree of the highest level in which they enrol within 9 years of Year 10. Results are weighted by the number of students in Year 10 in the TTWA in 2002. Standard errors are two-way clustered by base year TTWA.

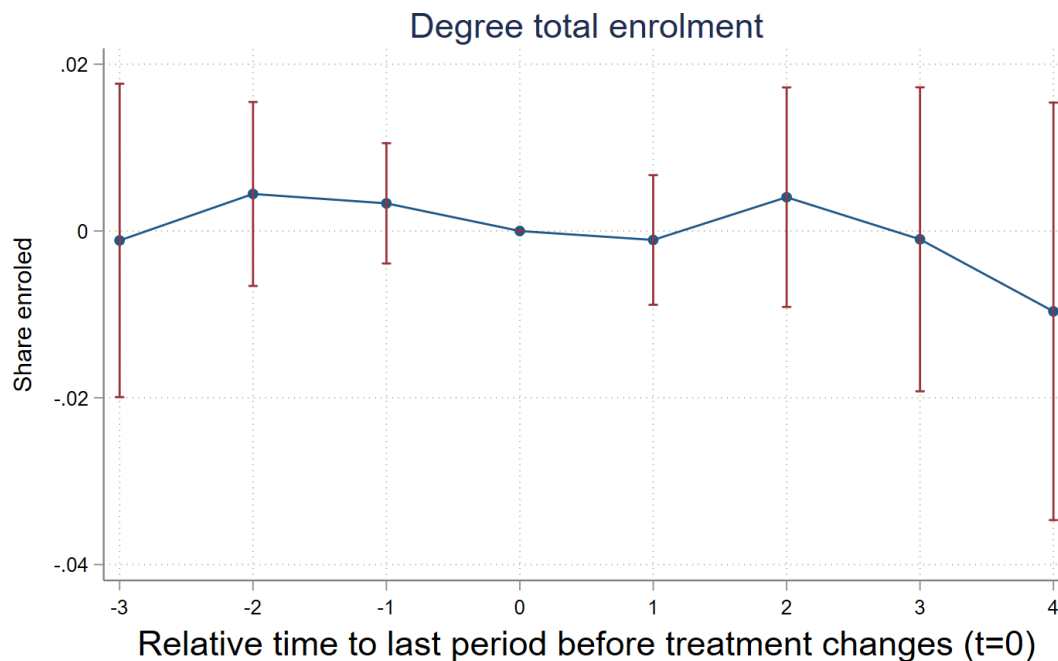
Table 2: Relationship between local field share and student subject choice

	A-Level field	Uni Field	A-Level Field	Uni Field
Share field emp.	0.107 (0.149)	0.316*** (0.0390)		
Δ_5 Share field emp.			-0.523* (0.238)	-0.0835 (0.111)
Field x Year FE	Y	Y	Y	Y
N	15345	18600	8370	10850

Data at the TTWA x Year x Qualification level. Employment shares and changes from the BSD. Student information from the NPD. Cohorts that were in Year 10 from 2004-2013 are included. Student information for HE enrolment comes from HESA. For university enrolment, cohorts that were in Year 10 from 2005-2012 are included. Students are allocated to the first degree of the highest level in which they enrol within 9 years of Year 10. Results are weighted by the number of students in Year 10 in the TTWA in 2002. Standard errors are two-way clustered by base year TTWA.

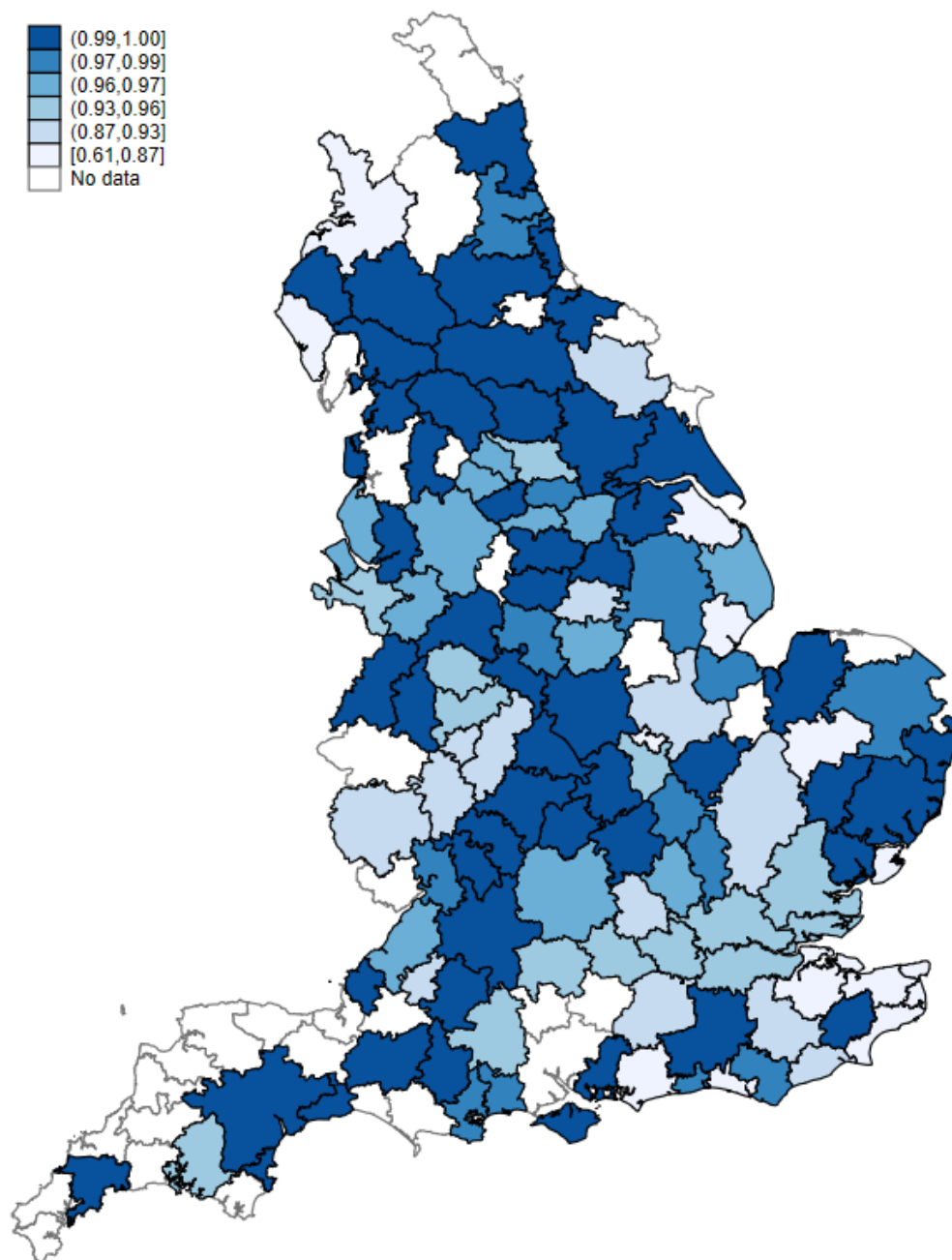
A Appendix - Results

Figure A1: Degree enrolment, total economic conditions



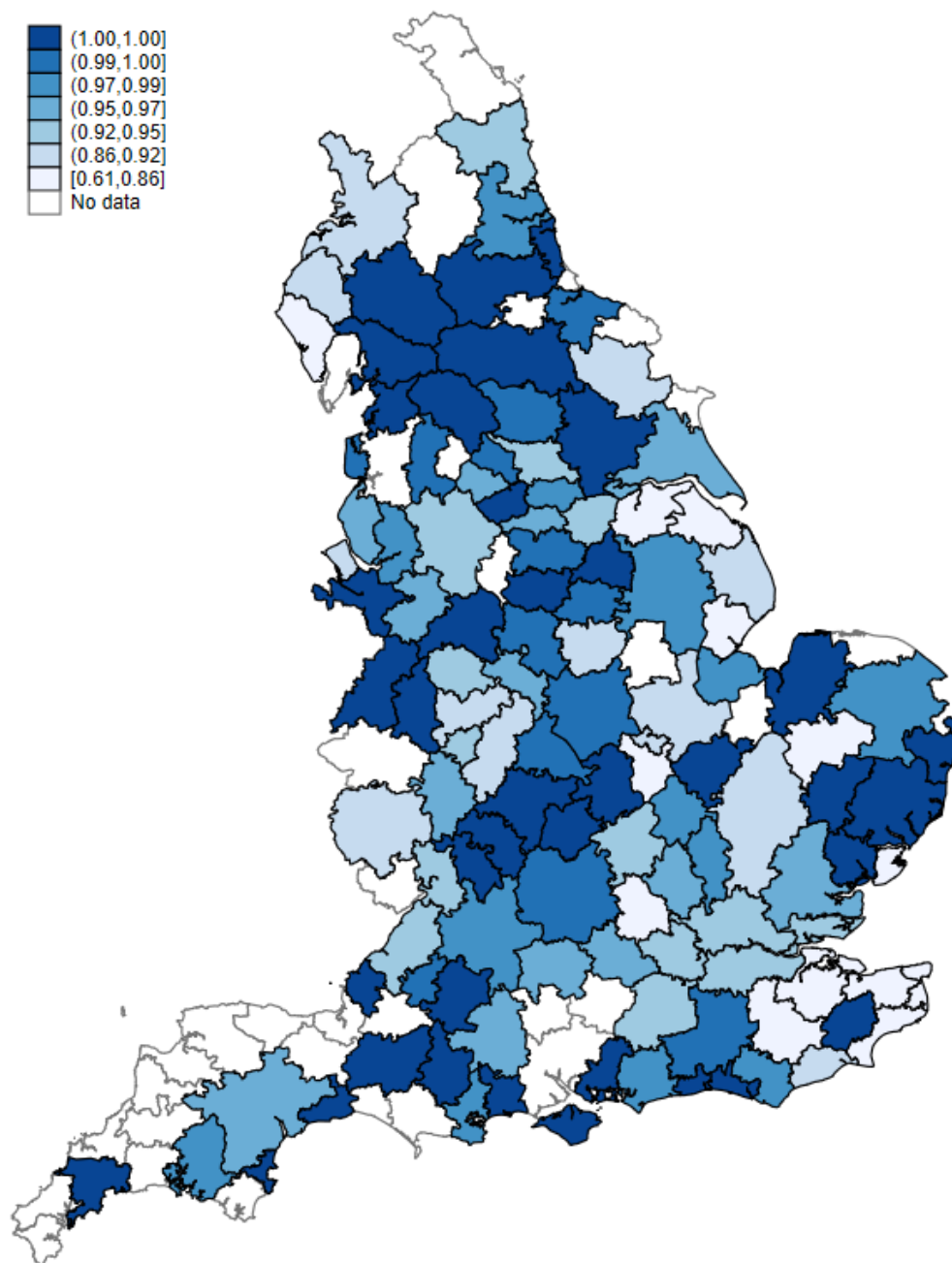
Data at the TTWA x Year level. Employment shares and changes from the BSD. Student information for HE enrolment comes from HESA. For university enrolment, cohorts that were in Year 10 from 2005-2012 are included. Students are allocated to the first degree of the highest level in which they enrol within 9 years of Year 10. Standard errors are two-way clustered by base year TTWA.

Figure A2: Student-weighted share of schools offering A-level biology



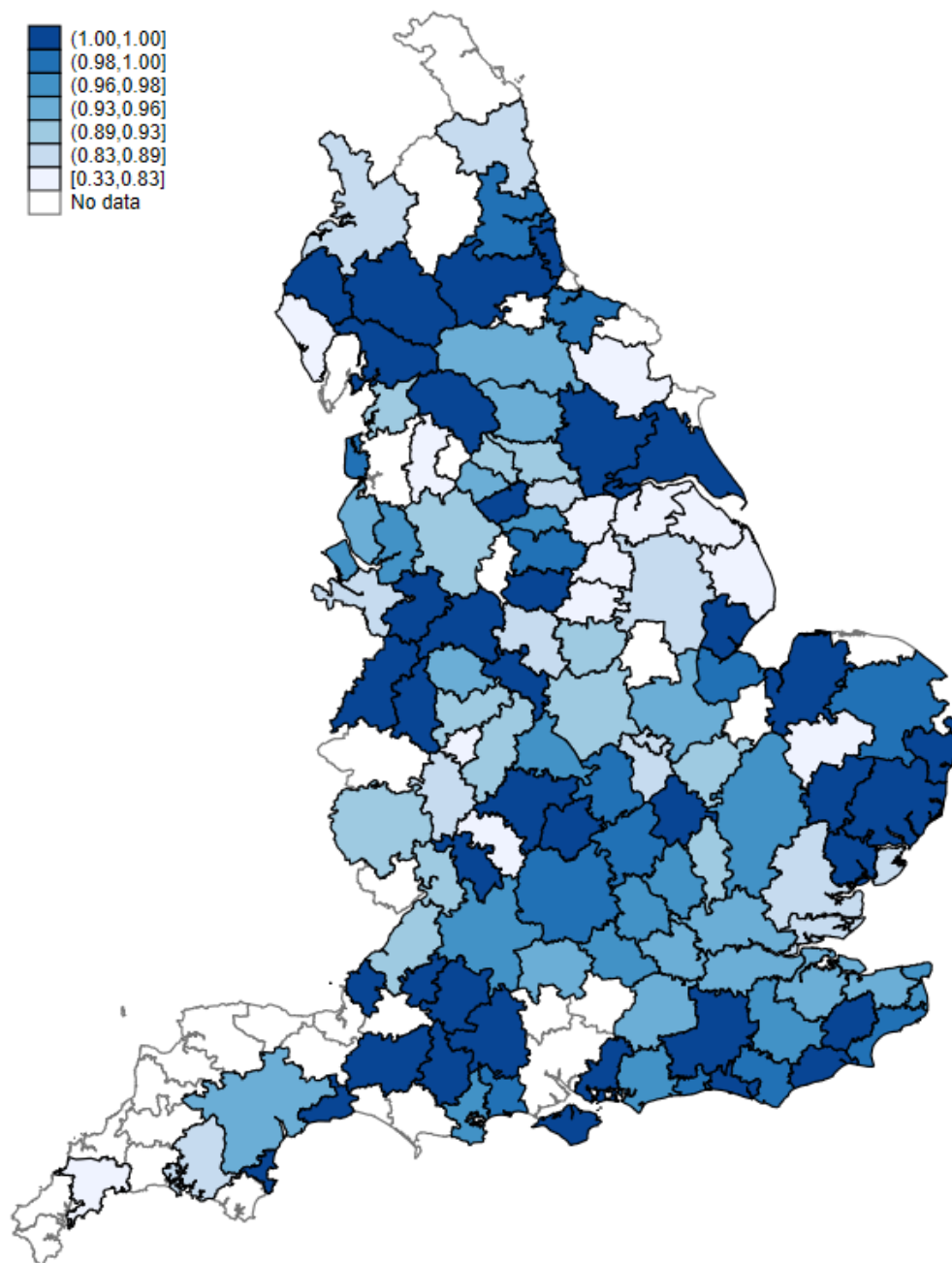
For each TTWA, the map shows the student-weighted share of schools which, in 2009 offered A-level biology.

Figure A3: Student-weighted share of schools offering A-level chemistry



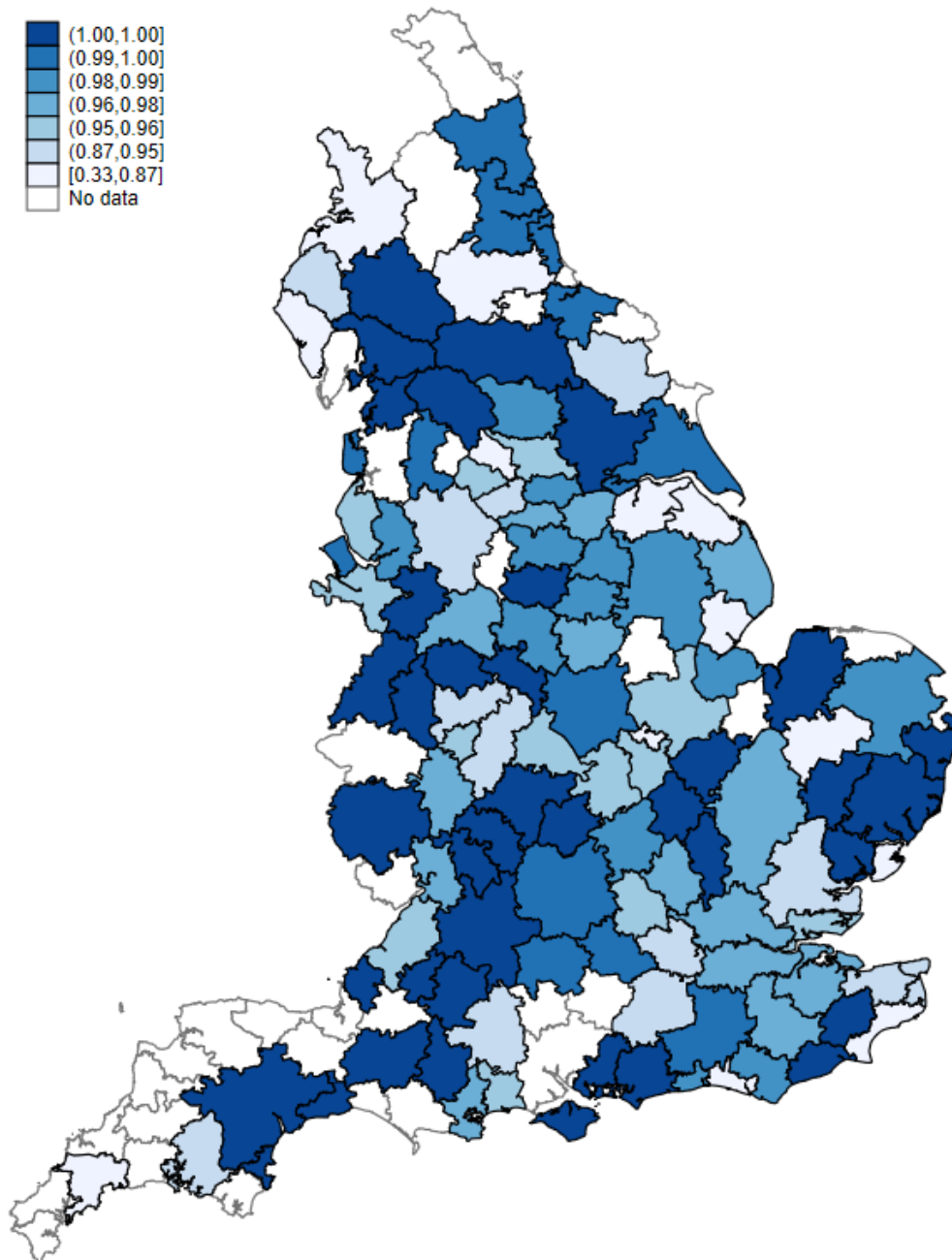
For each TTWA, the map shows the student-weighted share of schools which, in 2009 offered A-level chemistry.

Figure A4: Student-weighted share of schools offering A-level english literature



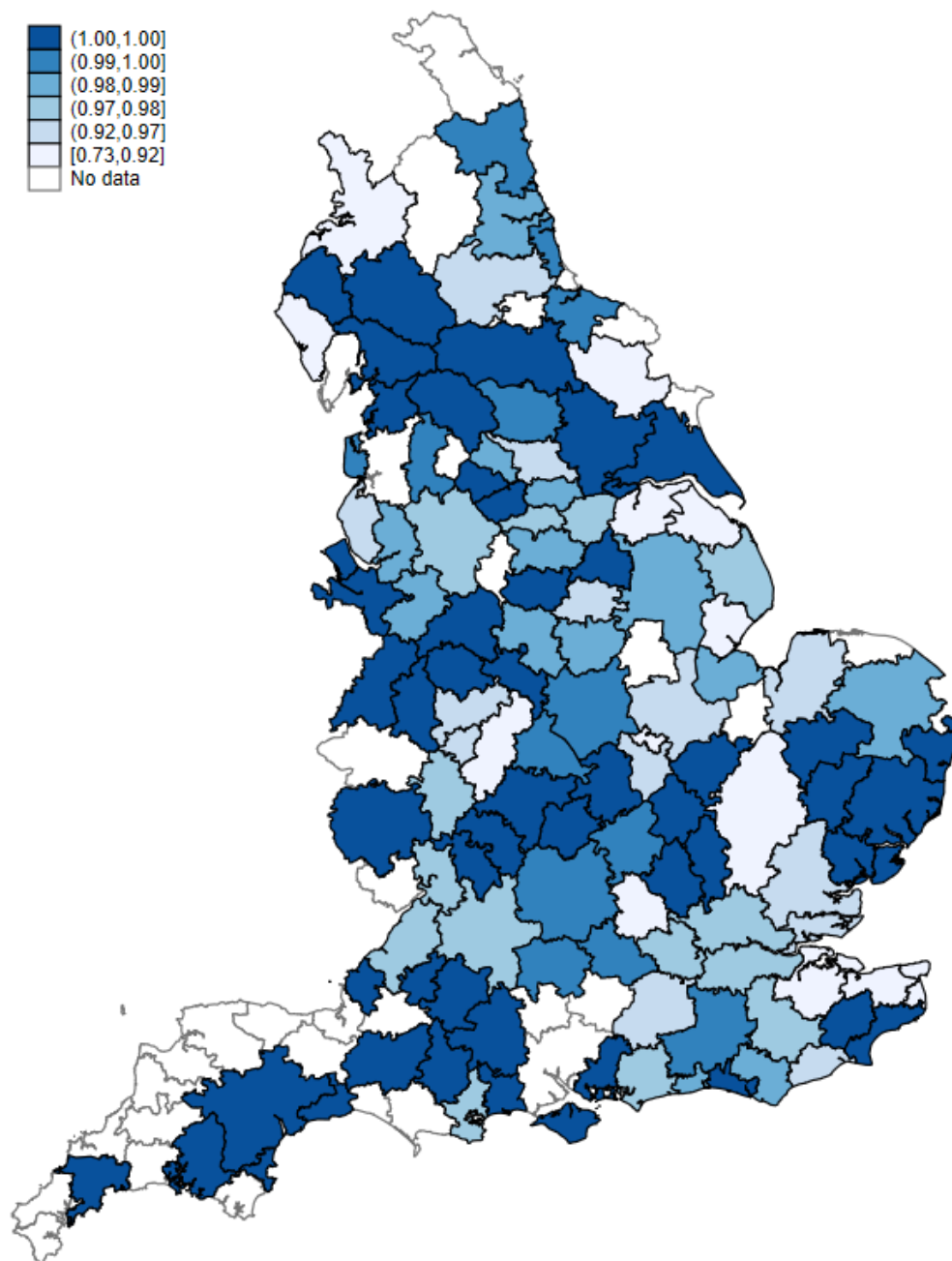
For each TTWA, the map shows the student-weighted share of schools which, in 2009 offered A-level English literature.

Figure A5: Student-weighted share of schools offering A-level history



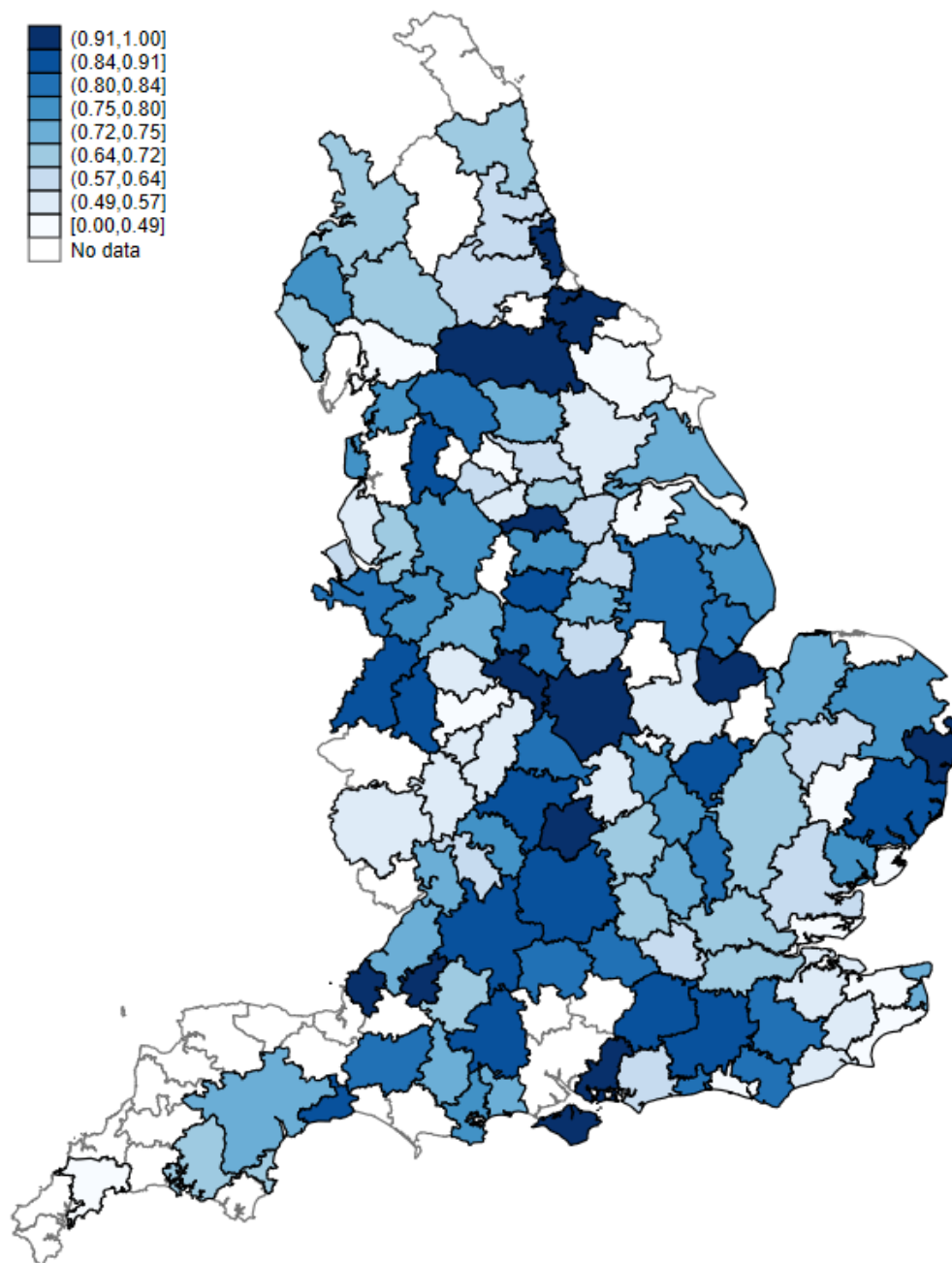
For each TTWA, the map shows the student-weighted share of schools which, in 2009 offered A-level history.

Figure A6: Student-weighted share of schools offering A-level mathematics



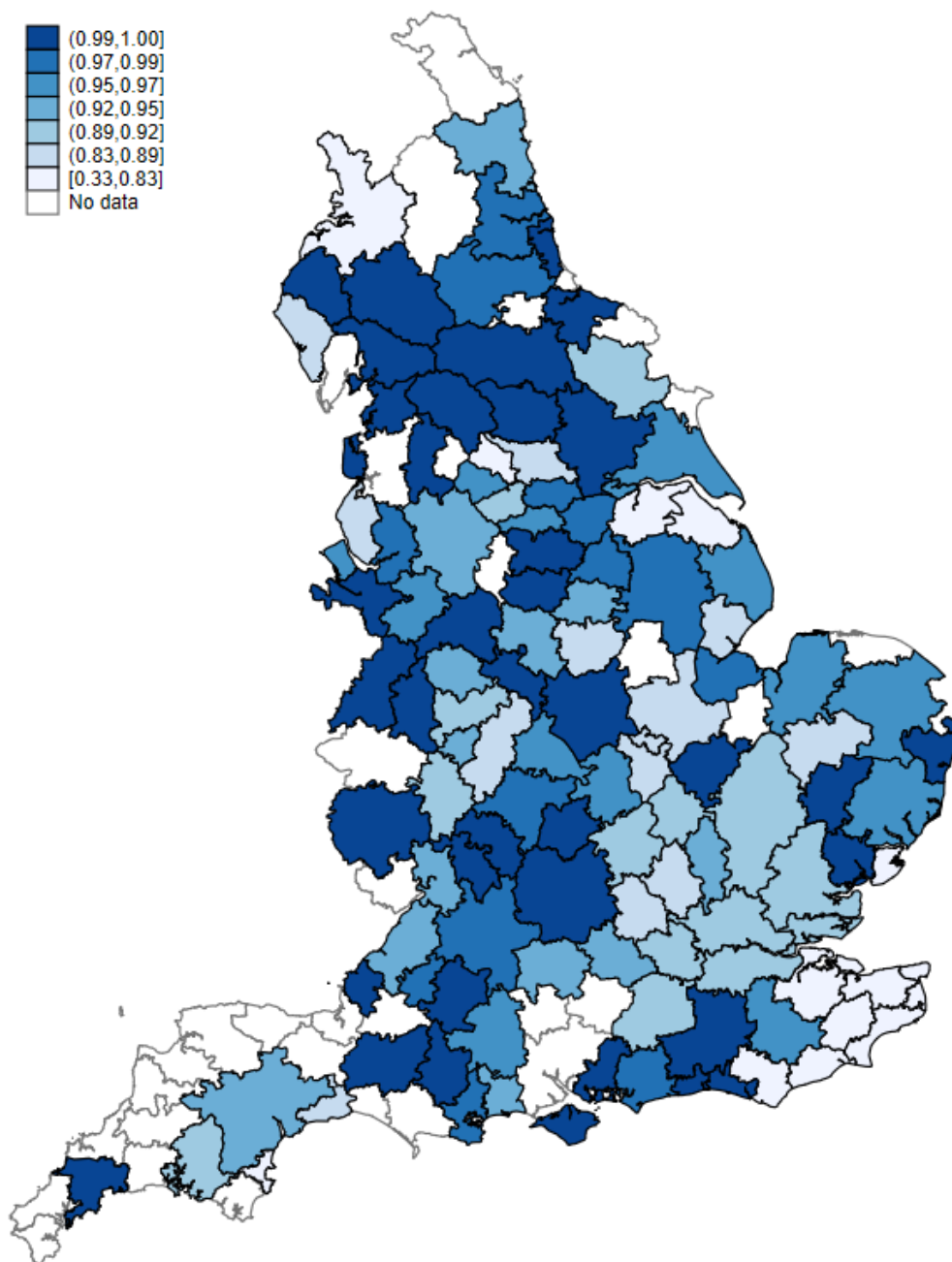
For each TTWA, the map shows the student-weighted share of schools which, in 2009 offered A-level mathematics.

Figure A7: Student-weighted share of schools offering A-level further mathematics



For each TTWA, the map shows the student-weighted share of schools which, in 2009 offered A-level further mathematics.

Figure A8: Student-weighted share of schools offering A-level physics



For each TTWA, the map shows the student-weighted share of schools which, in 2009 offered A-level physics.

Table A1: Relationship between local total employment and student degree enrolment

	A-Level Enrol.	A-Level Passes	Uni Enrol.	A-Level Enrol.	A-Level passes	Uni Enrol.
Indexed total emp.	-0.0657** (0.0247)	-0.0738 (0.0962)	-0.0590* (0.0260)			
Δ_5 Indexed total emp.				-0.0534 (0.0293)	-0.0894 (0.0838)	-0.0419 (0.0268)
Year FE	Y	Y	Y	Y	Y	Y
N	1705	1705	1705	930	930	930

Data at the TTWA x Year level. Employment shares and changes from the BSD. Student information from the NPD. Cohorts that were in Year 10 from 2004-2013 are included. Student information for HE enrolment comes from HESA. For university enrolment, cohorts that were in Year 10 from 2005-2012 are included. Students are allocated to the first degree of the highest level in which they enrol within 9 years of Year 10.

B Appendix - Data Sources

B.1 Business Structure Database (BSD)

To observe local variation in skill demand, I use the ONS Business Structure Database (BSD), an annual snapshot of the Inter-Departmental Business Register (IDBR), which is a live register of data collected by HM Revenue and Customs via VAT and Pay as You Earn (PAYE) records. This covers all businesses liable for VAT (turnover exceeds the VAT threshold) and/or has at least one member of staff registered for the PAYE tax collection system, then the business will appear on the IDBR (and hence in the BSD). My primary unit in this data is the Local Unit (establishment), which is tied to a specific Enterprise (firm) in each year. I use the location, industry, and employment for each establishment from 2002-2019 to compute the local industrial composition and map that industrial composition to demand for skills using the LFS, as described below.

B.2 Student Data

Data on students and education choices comes from several datasets provided by the Department for Education (DfE). The main source of data, from which I use information on students' geographic location in the specified base year, is the National Pupil Database (NPD). This is a linked administrative dataset of all students in state-funded schools in England. I observe students' location, demographic characteristics, school information, and exam results during the period of compulsory education. I also use data from the Higher Education Statistics Agency (HESA) dataset, which covers students registered at a higher education provider in a course that leads to a qualification at level 3 or higher.

B.3 Labour Force Survey (LFS)

I use the UK Labour Force Survey (Office For National Statistics, 2023) to map from industries to skills. To define the skill composition, I harmonize the EHATFLD (field) and HIQUAL (level) variables over time, and populate them for earlier years using the given information. The data used is for those ages 18 to 34, employed, in years 2004-2008, quarterly. The field of education is assigned for those who completed a qualification at Level 3 or higher. While A-Levels are a level 3 qualification in the qualifications framework, field is not assigned for students taking A-levels.

I consider all qualifications for which a field is assigned; specifically the field shares include those in both Level 3+ vocational qualifications, which results in the skill mix being somewhat different than those typically taken in academic qualifications. I do this as the fields studied in vocational qualifications are still indicative of the relevant academic qualifications.

The fields and level of qualifications I consider are listed below:

C Appendix - Data Processing

C.1 Mapping Fields

After using the LFS to map the fields for workers, I standardize the DfE and HESA data to the same fields. A-Level qualifications are assigned a field based on the sector / subject framework values assigned to the qualification in the Qualification Accreditation framework. These values provide detailed description of the subject studied (e.g. Accounting and Finance; Transportation Operations and Maintenance; Hospitality and Catering), which allows me to map them to the standardized fields used in the LFS. For example, A-levels in economics are mapped to 0300 - Social sciences, journalism and information, and biology is mapped to 0500 - Natural sciences, mathematics and statistics.

University degrees are assigned a subject based on the Joint Academic Coding System (JACS) to match the assignments in the LFS. The exact mapping used in the LFS is spelled out in the data dictionary, and I use the JACS3 codes provided in the HESA data to map degrees (by title) accordingly.

Code	Label
0000	Generic programmes and qualifications
0100	Education
0200	Arts and humanities
0300	Social sciences, journalism and information
0400	Business, administration and law
0500	Natural sciences, mathematics and statistics
0600	Information and communication technologies
0700	Engineering, manufacturing and construction
0800	Agriculture, forestry, fisheries and veterinary
0900	Health and welfare
1000	Services

Table B2: Qualification Fields

Code	Label
0	Other
1	Degree or equivalent (Level 6 or above)

Table B3: Qualification Levels

C.2 BSD Smoothing

Despite being an administrative database, the BSD has some gaps in reporting. To avoid erroneous interpretation of missing data as a decrease and immediate increase in industry employment (and therefore the associated skills), I use the imputation procedure described below. For firms with a gap in reporting between two years in which they are active, I assume their location and industry stays the same as in the initial year. I linearly impute employment between the start and end year, as in the example below.

Year	Employment	Imputed year	Imputed year
2002	210	2002	210
		2003	220
		2004	230
2005	240		

Table C4: Imputation example. The location and industry from the 2002 record would be used for the 2003 and 2004